# FastSpeech 2 for English and Telugu

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Technical report: implementation, data, training and analysis of two separately trained FastSpeech 2 models (English: Svarah; Telugu: IndicTTS Telugu) · 7 November 2026
Code, results and checkpoints: `https://github.com/ashdev-7/iiith-fastspeech2`

**Sources of numbers.** Every quantitative claim is labelled: paper reported in the FastSpeech 2 paper (Ren et al., ICLR 2021; LJSpeech); repo taken from the reference implementation `ming024/FastSpeech2` (commit d4e79eb) or its configs; measured measured in this project (Kaggle jobs, each logging its results and the commit it ran in `status.json`; the jobs are listed in `kaggle/README.md`); estimated an interpretation, not a measurement. Intelligibility is measured by automatic speech recognition (Whisper large-v3 for both languages; for Telugu, transcripts that Whisper wrote in Kannada script are also scored after mapping them back to Telugu, §5.5), pinned to exact revisions; no formal listening test (MOS) was run. Sections marked beyond the brief go further than the assignment asks.

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### Results at a glance

| | English (Svarah) | Telugu (IndicTTS) |
|---|---|---|
| Data used, train / validation | 4.87 h / 0.98 h; 110 / 109 speakers measured | 4.90 h / 1.01 h; 2 speakers measured |
| Training | 17,949 steps, 3.5 h on 1 × T4 | 55,524 steps, 8 h on 2 × T4 |
| Validation loss (total), best → final | 2.651 (7k) → 2.686 | 1.970 (16k) → 1.974 |
| Validation duration MAE / Pearson r (log) | 2.81 frames / 0.58 | 1.06 frames / 0.85 |
| Validation pitch RMSE / r (phone level) | 34.6 Hz / 0.35 | 29.2 Hz / 0.76 |
| ASR character error, validation synthesis (copy-synthesis ceiling) | 15.7 % (11.6 %), Whisper large-v3 | 35.8 % (33.1 %), Whisper large-v3, 150 utt.; weak on Telugu (§7) |
| ASR character error on the five test sentences (corpus) | 3.6 % (final configuration, Whisper large-v3, normalised) | 27.1 % (Whisper large-v3, script-mapped) mostly script and digit formatting by the ASR (§5.5) |
| Real-time factor (acoustic + vocoder), T4 / 4-thread CPU | 0.027 / 0.35 | 0.025 / 0.38 |
| Required outputs | 10 wavs, 30 variance plots, training- and validation-loss plots for both languages; every test sentence passes the run-time checks (Σ durations = mel length, no zero-length phones, finite non-silent audio) measured. Checklist: Appendix. | |

Beyond the brief: an evaluation on every held-out utterance (§7); a diagnosis of where quality is lost and three remedies tested (§5.1–5.3); a measured check that the Telugu aligner's access to the validation clips does not bias the results (§2.2); 36 further test sentences (§7.3–7.4); 12 fixes to the reference code (§1.3); over 100 automated tests run before every compute job (§1.4).

# 1. Implementation approach

## 1.1 Architecture used from the reference implementation

The starting point is `ming024/FastSpeech2` (MIT licence), vendored with the modifications in §1.3. The model is the non-autoregressive FastSpeech 2 acoustic model:

- **Encoder**: phoneme embedding + 4 feed-forward Transformer (FFT) blocks (self-attention, 2 heads, hidden 256; Conv1D 1024, kernels 9/1).
- **Speaker embedding** (multi-speaker mode) added to the encoder output: 110 Svarah training speakers, 2 Telugu speakers.
- **Variance adaptor**: three predictors (2 × Conv1D 256, kernel 3, LayerNorm, dropout 0.5) for log-duration, pitch and energy. Pitch and energy are predicted *per phone*, quantised into 256 bins, embedded and added to the hidden sequence; the **length regulator** repeats each phone vector $d$ times, with $d$ = round(exp(log $\hat{d}$) − 1) at inference.
- **Decoder**: 6 FFT blocks → linear to 80 mel bins → 5-layer convolutional **PostNet** (residual).
- **Loss**: MAE on mel (before and after PostNet) + MSE on log-duration, pitch and energy (6 terms logged).
- **Vocoder**: the reference repository's pretrained **HiFi-GAN "universal"** generator (not trained here).

| Aspect | Paper (LJSpeech) paper | Reference repo repo | This project |
|---|---|---|---|
| Data | LJSpeech, 24 h, 1 speaker | LJSpeech / AISHELL-3 / LibriTTS configs | ~5 h per language; 110 / 2 speakers |
| Pitch target | frame-level F0 via continuous wavelet transform | phone-level average F0 (FastPitch-style), no CWT | as repo |
| Decoder layers / dropout | 4 / 0.1 | 6 / 0.2 | as repo |
| Batch / steps | 48 / 160k | 16 / 900k (LJSpeech config) | 16 / 18k (en), 56k (te), sized from measured throughput |
| Vocoder | Parallel WaveGAN | HiFi-GAN / MelGAN | HiFi-GAN universal |
| Reported quality / speed | MOS 3.83 ± 0.08; RTF $1.95 \times 10^{-2}$ on a V100 (incl. vocoder) | — | RTF 2.5–2.7 × $10^{-2}$ on a T4 measured; no MOS |

The paper's MOS and RTF cannot be compared directly with ours: different data (24 h single speaker vs 5 h), vocoder, GPU, and no listening test here.

## 1.2 Language-specific work

### Telugu (the reference has no Telugu support)

- **Grapheme-to-phone** (`text/telugu/g2p.py`, `phones.py`): Telugu script is close to phonemic, so it is converted by deterministic rules on the Unicode Telugu block: a consonant carries the inherent vowel /a/ unless a vowel sign or virama follows; a vowel sign replaces it; the anusvara (ం) becomes the nasal homorganic with the next consonant (బలంగా → `b a l a ng g aa`, ఉంచండి → `u ny c a nn dd i`), otherwise /m/; visarga → /h/. 50 phone labels in ASCII (retroflex consonants doubled: `tt dd nn ss ll`; aspirates with `h`; long vowels doubled). Every code point must be consumed by a rule, otherwise an error is raised, so characters are never dropped silently.
- **Normalisation** (`text/telugu/normalize.py`): Telugu number words for cardinals, ordinals and times: 2026 → రెండు వేల ఇరవై ఆరు, 12వ → పన్నెండవ, 10:45 → పది నలభై ఐదు.
- **Symbol table**: a separate 65-symbol Telugu table (padding, punctuation, 50 phones, silences); the encoder embedding is sized per language (L1–L4 in `MODIFICATIONS.md`, which also has a one-table summary of all Telugu work).
- **Alignment**: no pretrained Telugu MFA model exists, so an MFA acoustic model was **trained here** on the selected IndicTTS clips with a dictionary generated by the G2P (11,947 words, 0 without pronunciation) measured; §2.2 shows that its access to the validation clips does not affect the results.

- **Code-mixing** (beyond the test set): English words written in Latin script inside Telugu text (e.g. "iPhone") are mapped from ARPAbet to the nearest Telugu phones, using only phones already in the Telugu table (`text/telugu/latin.py`).
- **Plots**: Telugu titles are shaped correctly (Pillow with raqm and Noto Sans Telugu), so no glyphs are broken.

**English**

- **Inference frontend** (`text/english_frontend.py`): times ("10:45 AM" → "ten forty five A M", AM/PM as letter names), numbers and years (reference `inflect` rules: 2026 → "twenty twenty six"); homographs known to g2p_en (*conduct*) resolved by part of speech over the whole sentence; pronunciations from the LibriSpeech lexicon (the lexicon used for alignment); g2p_en for out-of-vocabulary words; punctuation → short pause `sp`. Every decision is recorded per word in `frontend_trace.json`; the test text itself is never changed.
- **Rules found by inspecting the test outputs** (§5.4), all general, none word-specific: split words ("rhyth  m") are joined for pronunciation when the concatenation is a lexicon word and the first part is not; a word-final "oo" in an out-of-vocabulary word is /uː/ (*andaroo*); the **citation form** is preferred where the lexicon's first variant is a casual reduction (dropped /t/ in "thousand", "twenty", "parliamentary"; flapped /t/ in "thirty"; 670 words in all); "the" before a vowel is /ðiː/; and an out-of-vocabulary word of 12 or more letters is split into the fewest lexicon words of at least 4 letters (*antidisestablishmentarianism* → anti + disestablishment + arianism), instead of being left to g2p_en.
- **Robustness rules for unseen text** (from probing with new sentences; none changes the ten test sentences, checked phone by phone): "%" and "&" spoken; an hour with a.m./p.m. and no minutes; acronyms spelled or taken from CMUdict; "1,247" read as a quantity, not a year; "Month day" dates as ordinals; no pause after "Dr."; two corrections to g2p_en's homograph table (past-tense *read*, verb *process*).
- **Alignment**: MFA's pretrained `english_us_arpa` acoustic model with the LibriSpeech lexicon + g2p_en for out-of-vocabulary words (6,388 words, 0 without pronunciation) measured.
- **Inference-time options** (§5.2–5.3): a global-variance post-filter and clear-speech options, each introduced for a measured problem; the trained weights are the same for every output.

### 1.3 Modifications to the reference code

Twelve fixes (M1–M12) and four language-support changes (L1–L4) are listed in `MODIFICATIONS.md` with the original behaviour, the reason, the new behaviour and the test that covers it. The ones that change results:

| # | Problem in the reference | Fix |
|---|---|---|
| M1 | Energy embedding received the *pitch* control value | uses `e_control` |
| M5 | Utterances without a TextGrid added the previous utterance's pitch/energy to the normalisation statistics | statistics only from processed utterances |
| M6 | Empty TextGrid intervals (MFA ≥ 2 writes pauses this way) were dropped, so pauses vanished from phone sequences and durations | kept as `sp` |
| M9 | float64 energy targets: `backward()` failed ("Found dtype Double") | cast to float32 |
| M10 | numpy-scalar learning rate in checkpoints, rejected by `torch.load(weights_only=True)` | `float()` |
| M12 | Pitch/energy controls multiplied the *z-normalised* value: for a voice below the corpus mean, pitch × 1.2 *lowered* F0 (126 → 113 Hz) measured | scale in Hz / energy units; now × 1.2 → 126 → 157 Hz (en), 209 → 252 Hz (te) measured |

Others: CPU-safe STFT (M2), librosa ≥ 0.10 keyword API (M3), UTF-8 file handling (M4, M7), sorted speaker ids (M7), `map_location`/`weights_only` loading and HiFi-GAN paths (M8), masks on the input's device (M11).

### 1.4 Engineering for a reliable run

All compute ran as Kaggle script jobs that clone the repository and record the commit. Each job first runs the test suite (over 100 tests: frontends, symbol tables, split leakage, preprocessing, model forward/backward on every test sentence, checkpoint round trip, plots, prosody and clarity rules); a CPU **rehearsal** ran every stage on a small real

subset before the full runs. The training job measures throughput for 300 steps, sizes the schedule to a time budget, and runs a **dry run of inference and plotting** on that 300-step checkpoint before the long training. A **watchdog** kills a training process that exceeds its budget or stops logging; it was added after the first English run hung inside training (two-GPU DataParallel) and was killed at Kaggle's 12-hour limit with no checkpoint. The rerun used one GPU and completed.

## 2. Data: selection, split and preprocessing

### 2.1 Selection and subsampling

Both datasets are larger than the 5 h + 1 h budget, so a deterministic, duration-targeted selection was used (`data/select_split.py`). Only the two specified datasets are used, from Hugging Face at pinned revisions (`ai4bharat/Svarah`, `SPRINGLab/IndicTTS_Telugu`).

- **Unit of assignment = group**, never a single clip: a Svarah *recording* (all chunks of one recording), or for Telugu all utterances with the same normalised transcript (both IndicTTS speakers often read the same sentence). Groups that share a transcript are additionally merged (union-find), so no sentence can occur in both splits.
- **Order**: groups are visited in a seeded pseudo-random order (SHA-1 of seed and group id), independent of file order.
- **Quotas**: each speaker gets train/val hours proportional to its share of the candidate audio; a group goes to validation if it fits all validation quotas it touches, else to training if it fits, else it is not used.
- **No overlap**: utterance ids, groups, normalised transcripts and audio content (SHA-1 of the PCM samples) must be disjoint between train and validation; duplicate audio is excluded. All checks passed (0 overlaps) measured.

| | English (Svarah) | Telugu (IndicTTS) |
|---|---|---|
| Selected train | 3,079 utt. · 4.87 h · 110 speakers | 2,164 utt. · 4.96 h (female 2.41 h, male 2.55 h) |
| Selected validation | 742 utt. · 0.98 h · 109 speakers | 450 utt. · 1.02 h |
| Not aligned by MFA | 0 | 28 (1.1 %), in 4 clusters of consecutive ids |
| Rejected in preprocessing | 2 (≤ 1 voiced frame) | 0 |
| Used for training / validation | 3,077 / 742 | 2,140 / 446 |
| Hours used, train / validation (selected audio) | 4.87 / 0.98 h | 4.90 / 1.01 h |
| Same, after leading/trailing silence trimming (mel frames) | 4.53 / 0.92 h | 4.21 / 0.87 h |
| Overlap of the used sets: ids / transcripts | 0 / 0 | 0 / 0 |

Hours and overlaps of the used sets: Kaggle `fs2-13-data-accounting`, `Evaluation/Data/accounting.json`. The unaligned Telugu clips are clustered (ids te_0091x–0092x, te_0233x, te_0264x), which suggests transcript/audio mismatches in those ranges rather than random failures estimated; they were excluded, not repaired. Svarah validation speakers also appear in training (different recordings): the split is by recording, not by speaker, so the English validation set measures generalisation to new sentences, not to new voices.

### 2.2 Preprocessing

1. **Audio**: decoded from the Hugging Face parquet files, mono, resampled to 22,050 Hz (soxr), 16-bit.
2. **Text**: English transcripts normalised by the English frontend rules for the MFA dictionary; Telugu transcripts by the Telugu normaliser; one pronunciation dictionary per language generated from the project's own frontends, so training and inference use the same phone sets.
3. **Forced alignment** (MFA 3.4.3): English 3,821 / 3,821 utterances aligned; Telugu 2,586 / 2,614 measured. The English aligner is MFA's pretrained model and saw none of our data. The Telugu aligner had to be trained here; it was trained on the selected training and validation clips (no other audio), because the aligner also produces the validation duration targets. The FastSpeech 2 model itself trains only on the training split. To measure whether the aligner's access to the validation clips matters, a second Telugu aligner was trained on the **training clips**

**only** (same dictionary, same MFA version and command), the validation clips were aligned with it, and both alignments were compared (`tools/aligner_check.py`, Kaggle `fs2-18-aligner-check`, `Evaluation/Data/aligner_check.json`) measured:

| Second aligner (training clips only) vs the one used | Validation clips (445) | Training clips (2,139) |
|---|---|---|
| Phones whose duration is within 1 frame (11.6 ms) / identical | 95.4 % / 66.8 % | 96.1 % / 69.1 % |
| Mean phone-boundary shift (95th percentile) | 2.7 ms (10 ms) | 2.6 ms (10 ms) |
| Clips with the same phone sequence (the others differ only in where a short pause was placed) | 363 | 1,951 |

The released model's validation duration error against the two alignments: **1.058 → 1.112 frames** (MAE) and Pearson r of log durations **0.848 → 0.823**. The reported Telugu duration scores are therefore optimistic by about 0.05 frames (0.6 ms per phone), and the training targets the model learned from would have been almost the same without the validation clips (agreement on the training clips is as high as on the validation clips).

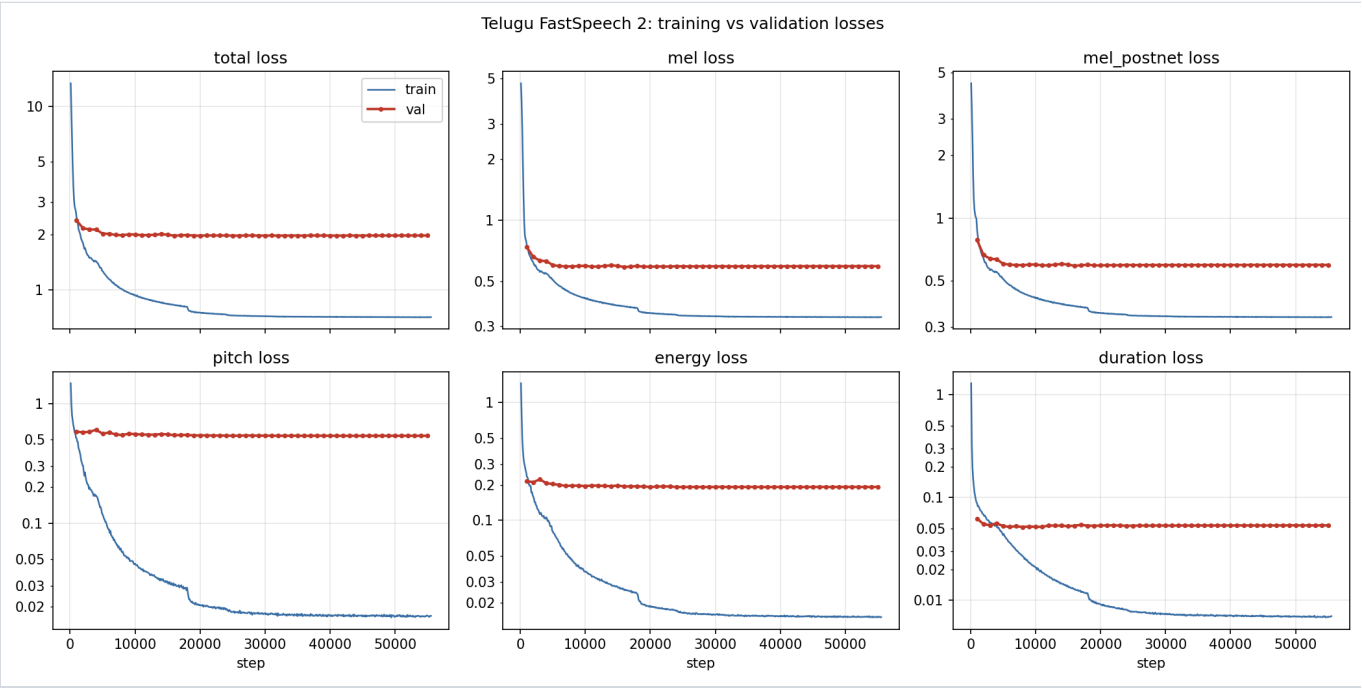
4. **Features** (reference preprocessor with fixes M5–M7): leading/trailing silence trimmed using the alignment; 80-band log-mel (FFT 1024, hop 256 = 11.6 ms, 0–8 kHz, matching the HiFi-GAN vocoder); F0 with WORLD DIO + StoneMask, unvoiced frames linearly interpolated, averaged per phone; energy = L2 norm of each STFT frame, averaged per phone; pitch and energy z-normalised with the training-set mean/std; per-phone durations in frames from the TextGrids.

## 3. Training configuration

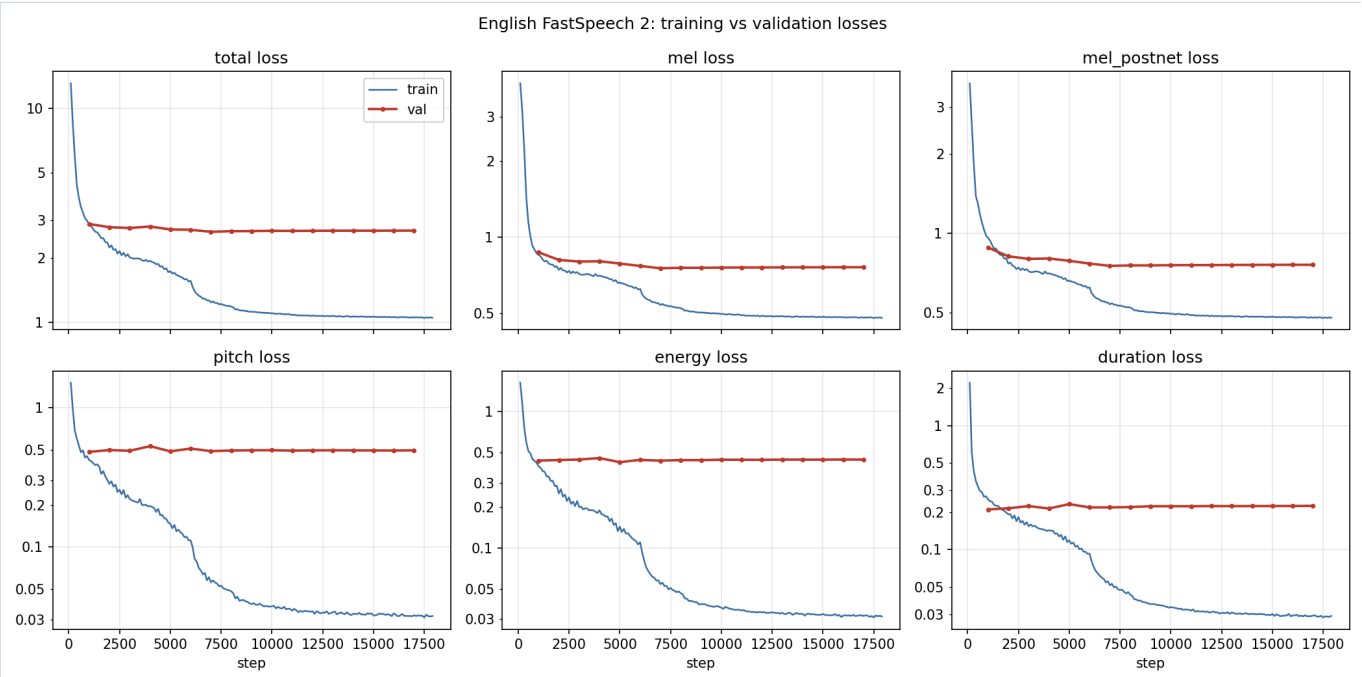
| | English | Telugu |
|---|---|---|
| Model | reference architecture (§1.1); 35.44 M (en) / 35.34 M (te) parameters, the difference being the symbol tables (360 vs 65 symbols) | |
| Optimiser | Adam, β = (0.9, 0.98), ε = $10^{-9}$; Noam warm-up 4,000 steps; ×0.3 decay at 1/3, 4/9 and 5/9 of the planned steps (the reference's 300k/400k/500k of 900k); gradient clipping 1.0; batch 16 (length-sorted groups of 4 batches); fp32 | |
| Planned steps / LR decays | 20,000 / 6k, 8k, 11k | 56,000 / 18k, 24k, 31k |
| How the plan was set | 300-step probe: 1.45 steps/s on 1 × T4; 210-min budget | 300-step probe: 2.08 steps/s on 2 × T4; 480-min budget |
| Completed | 17,949 steps in 211 min (stopped by budget) | 55,524 steps in 480 min (stopped by budget) |
| Validation | every 1,000 steps on the full validation set: 6 losses + duration MAE and total-length error (`val_diagnostics.csv`) | |

The English budget was cut from 8 h to 3.5 h for the rerun because the Telugu validation losses had stopped improving after ~8–10k steps (§4); this also saved GPU quota and shortened the exposure to another session failure. Exact configs: `English/config.yaml`, `Telugu/config.yaml`, and as run: `Results/<Lang>/train_config_used.yaml`.

## 4. Loss curves: convergence, stability and over-fitting




Telugu: training (blue, mean of each 100 steps) and validation (red, every 1,000 steps), log scale. Separate `training_loss.png` and `validation_loss.png` are in `Results/Telugu/`.




English, same layout (`Results/English/`).

| Validation loss | English best (step) | English final | Telugu best (step) | Telugu final |
|---|---|---|---|---|
| total | 2.651 (7k) | 2.686 | 1.970 (16k) | 1.974 |
| mel after PostNet | 0.753 | 0.760 | 0.589 | 0.595 |
| pitch (z-units$^2$) | 0.483 | 0.495 | 0.537 | 0.538 |
| energy | 0.427 | 0.445 | 0.192 | 0.192 |
| log-duration | 0.211 | 0.226 | 0.052 | 0.054 |

| | | |
|---|---|---|
| final *training* total | 1.049 | 0.707 |

**Convergence.** Training losses fall fast for ~2k steps (warm-up), then steadily; visible steps in the training curves coincide with the learning-rate decays (Telugu at 18k, English at 6k), after which training loss flattens. Validation losses reach within 2 % of their final value by step 7k (Telugu) and 5k (English) and are flat afterwards measured.

**Stability.** No spikes, no divergence, no non-finite losses in either run (the trainer stops on any non-finite value; it never triggered).

**Over-fitting.** The gap between training and validation grows throughout: by the end the training pitch and energy losses are 30–40× lower than validation for Telugu, ~15× for English. This is memorisation of 5 h of data, mainly by the variance predictors. For Telugu it is benign: validation never got worse (final within 0.2 % of best). For English there is **mild over-fitting**: validation pitch, energy and duration losses drift upward by 3–7 % after 7k steps, and the total by 1.3 %. Early stopping on validation loss (here ~7–10k steps) would give a slightly better English model and save time; the released checkpoint is the last one estimated.

**English vs Telugu level.** English validation losses are higher in every term except pitch, consistent with the harder data: 110 speakers (≈ 2.7 min each), accented, more varied speaking style and recording conditions, against two studio-quality Telugu speakers with ~2.5 h each.

## 5. Generated speech, per test sentence

Audio: `Results/<Lang>/test_generated_audios/`. Voices: English `svarah_8a8e2655` (chosen by recording clarity, §5.1); Telugu `te_female` (lower HiFi-GAN copy-synthesis error than `te_male`: mel L1 0.378 vs 0.394 on 20 validation clips each) measured. Pronunciation is assessed by transcribing each wav with an ASR model and comparing it with the frontend's normalised text (character error rate, CER); for English the CER is also given after Whisper's English text normaliser, which removes pure formatting differences ("10:45" vs "ten forty five").

**Test text, as provided.** The ten sentences are taken character for character from the text layer of the assignment PDF (PyMuPDF), which matches the rendered page; they are stored with their Unicode code points and SHA-256 in `tests/test_sentences.json`, and every inference run refuses a sentence whose hash differs. Other text extractions of the same PDF (pypdf, the Word copy) detach or reorder Telugu vowel signs (e.g. "విశ్లేషే ించారు" for "విశ్లేషించారు"), so they were not used. Spacing in the given text is kept as is, including "rhyth  m" (E3) and "అడ  విలో" (T3); the frontends only decide how such text is pronounced, and `frontend_trace.json` records the original text next to the model input. For "rhyth  m" the English frontend joins the pieces (read literally, "m" becomes the letter name: the first full run said "rythe em", §5.4); for "అడ  విలో" no rule is needed, because the Telugu input has no word-boundary symbol and the split gives exactly the phones of అడవిలో.

### 5.1 Choice of the English test voice, and where quality is lost measured beyond the brief

The first English outputs (voice `svarah_71f88d2a`, chosen because it has the most training audio) sounded muffled. A Praat analysis (`tools/acoustic_analysis.py`) and a diagnostic job (`fastspeech2/diagnose_quality.py`, Kaggle `fs2-07-diagnose`, `Evaluation/Diagnosis/`) located the cause. For 24–30 validation clips of each test voice it compares the natural recording, copy-synthesis (ground-truth mel → HiFi-GAN), FastSpeech 2 with ground-truth duration/pitch/energy (oracle) and free-running FastSpeech 2, band by band (long-term spectrum, dB relative to the natural recording):

| Band (Hz) | 0–500 | 500–1k | 1k–2k | 2k–4k | 4k–6k | 6k–8k |
|---|---|---|---|---|---|---|
| English, copy − natural | −0.1 | +0.7 | +0.5 | +0.8 | +1.0 | +1.1 |
| English, free-running − natural | −0.3 | +3.1 | 0.0 | +3.1 | +3.9 | +1.9 |
| Telugu, copy − natural | 0.0 | +0.4 | −0.7 | +0.1 | +0.2 | 0.0 |
| Telugu, free-running − natural | +0.3 | −0.3 | −2.4 | −1.9 | −1.1 | −0.2 |

Neither the vocoder nor the acoustic model removes high-frequency energy: both reproduce the recording to within ~1 dB (copy) and ~4 dB (free-running) in every band, and HNR, jitter and shimmer of all four versions are within 1.5 dB / 0.4 % / 1.2 % of each other. The muffled sound was **the voice's own recordings**: ranking all 110 training speakers by the clarity of their recordings (1–4 kHz band share + HNR/2, three training clips each, `clarity_en.json`) put `svarah_71f88d2a` 108th (1–4 kHz band −23 dB). Svarah is crowd-sourced; recording conditions vary by speaker. The test sentences were rendered in the four clearest voices; `svarah_8a8e2655` (rank 2, 1–4 kHz band −9.7 dB) had the lowest jitter and shimmer of the four and is used for all English outputs. The checkpoint is unchanged; only the speaker embedding chosen at inference differs. Effect at the time (means over E1–E4): spectral centre of gravity 300 → 620 Hz, jitter 1.2 → 0.5 %, shimmer 11.1 → 9.0 %. The selection rule was changed after listening to the first outputs; it is an objective rule on training recordings, and the earlier outputs remain in the repository history (commit `8f5ae5f`).

### 5.2 Over-smoothing: diagnosis and three remedies measured beyond the brief

The band-energy comparison above cannot see whether spectral *detail* is preserved. Measuring the variance of each log-mel bin over time ("global variance", GV) and the mean frame-to-frame change (flux) on the same held-out clips shows the classic FastSpeech 2 weakness: the decoder, trained with an L1/L2 loss, predicts conditional averages. English free-running mels keep 80–88 % of the natural GV and ~80 % of its flux, *also with ground-truth duration, pitch and energy* (so it is the decoder, not the variance predictors); copy-synthesis keeps 97–101 %. Telugu keeps

93–98 %. With 110 speakers of ~2.7 min each, the English decoder averages over very different voices and recording conditions; with two studio speakers, Telugu does not.

**Remedy 1 — GV post-filter (adopted for English).** `fastspeech2/postfilter.py` rescales each mel bin's deviation from its utterance mean by $r_k^{s/2}$, where $r_k$ is the ratio of natural to predicted variance, estimated once on **training** clips of the test voice (`Results/English/gv_ratio.json`), as in statistical parametric synthesis (global-variance compensation). Inference only. Scored on held-out clips of the test voice (`Evaluation/Postfilter/`, Kaggle `fs2-08-postfilter`):

| English, strength s | GV / natural | flux / natural | HNR (dB) | shimmer (%) | MCD (DTW) |
|---|---|---|---|---|---|
| 0 (off) | 0.76 | 0.62 | 16.9 | 8.2 | 36.7 |
| 0.5 | 0.90 | 0.67 | 17.4 | 8.1 | 37.6 |
| 1.0 (used) | 1.04 | 0.71 | 17.6 | 7.9 | 38.6 |
| Telugu 0 → 1.0 | 0.97 → 1.00 | 0.81 → 0.81 | 18.0 → 17.9 | 8.0 → 7.6 | 29.7 → 29.9 |

The post-filter restores the variance completely and part of the temporal sharpness, and makes the voice slightly less breathy and rough; MCD rises slightly because it rewards averaged spectra (a known property of MCD). Telugu is not over-smoothed (r ≈ 1.02, no measurable change), so its outputs are left unfiltered.

**Remedy 2 — HiFi-GAN fine-tuning on ground-truth-aligned mels (tried, not adopted).** The vocoder was fine-tuned per language on FastSpeech 2's teacher-forced mels of the training set, paired with the recordings cut exactly as the preprocessor cuts them, with discriminators (MPD + MSD, from the original HiFi-GAN code) trained from scratch because the reference ships only the generator (`fastspeech2/vocoder_finetune.py`, Kaggle `fs2-09-vocoder-ft-*`; `Evaluation/VocoderFinetune/summary.json`).

| Universal → fine-tuned | val. mel L1 (GTA) | Whisper CER, synthesis | Whisper CER, copy-synthesis | other |
|---|---|---|---|---|
| English | 0.866 → 0.850 | 15.7 → 15.3 % | 11.6 → 12.3 % | GV ratio 0.91 → 0.87; copy-synthesis HNR 14.9 dB vs 17.1 natural |
| Telugu | 0.671 → 0.600 | 35.8 → 35.5 % | 33.1 → 31.0 % | GV ratio 1.00 → 0.94 |

The fine-tuned vocoders fit the training mels better but did not make the synthesis clearer and made some outputs worse, so the universal HiFi-GAN is kept for both languages. The likely reason is the short adversarial training with discriminators started from scratch (fine-tuning normally starts from trained discriminators and runs far longer) estimated.

**Remedy 3 — speaker adaptation of the decoder (tried, not adopted).** The trained model was fine-tuned on the test voice's 29 **training** clips only (encoder frozen), with the step chosen on its 9 validation clips (`fastspeech2/adapt_speaker.py`, Kaggle `fs2-11-adapt-en`, `Evaluation/Adaptation/`). The training mel loss fell from 0.47 to 0.38, but the held-out clips improved only up to step 100–150 and then got worse; the gains (≤ 0.5 MCD on 9 clips) are within noise and the voice-quality measures do not move: with 2.7 min of one voice the decoder memorises rather than adapts. The unadapted model is kept.

### 5.3 Clear-speech inference options (English) measured beyond the brief

Listening to the English outputs exposed three further problems. Each was measured, and each remedy acts at inference time (`fastspeech2/clarity.py`, `prosody_rules.py`); the model weights are unchanged. The model's own predictions are kept in every variance file and drawn in every plot (§6), so nothing here hides what the variance adaptor predicted.

- **Endings cut off.** Training clips are trimmed at the first and last phone, so the model's last frame is still at speech level and the waveform stops mid-sound (last 20 ms of the model-only outputs: -42 dBFS). Remedy: a short silence lead-in, a 5-frame fade of the last frame and trailing silence before the vocoder (-86 dBFS).

- **Background noise.** The vocoder alone is clean, but the model reproduces the Svarah recordings' noise floor in its pauses (43.8 dB above digital silence, measured on 119 pause frames of the model's own output on three neutral sentences that are not test sentences, `Results/English/noise_profile.json`). Remedy: subtraction of that floor in the mel domain. Quietest frames -46.6 → -54.2 dBFS; speech level -17.1 → -16.5 dBFS (unchanged) (`tools/clarity_measure.py` on E1–E4, model-only vs final; `Evaluation/English/clarity_levels.json`).
- **Fast, run-together speech.** On validation the model is 4 % shorter than the real speakers (median length ratio 0.96), and long words were run together with their neighbours. Remedy: overall duration × 1.10; words with four or more vowels × 1.25, preceded by a 70 ms micro-pause (placed before their article, never between article and noun); phrasing (§6: lengthening before punctuation, a minimum pause at each punctuation mark, a falling end for statements and a rising end for yes/no questions).

**Which combination.** Every configuration was rendered on Kaggle and scored with Whisper large-v3 on the five test sentences and 18 further sentences (§7.3–7.4); the table is `Evaluation/English/variants_summary.json`:

| English configuration (same checkpoint and voice) | CER, 5 test sentences | CER, all 23 | E4 |
|---|---|---|---|
| model + GV post-filter † | 2.7 % | 2.9 % | 2.9 % |
| + terminal fall † | 2.7 % | 3.2 % | 2.9 % |
| + fall + lexical stress | 6.9 % | 4.1 % | 17.1 % |
| + fall + stress, denoise, edge fade | 7.8 % | 4.0 % | 15.7 % |
| **final: fall + stress + phrasing, duration × 1.10, long words × 1.25 with micro-pauses, denoise, edge fade** | **3.6 %** | **2.3 %** | 17.1 % |
| final without the lexical-stress rule (Kaggle `fs2-21-final-en-nostress`) | 3.9 % | 2.3 % | 15.7 % |

† Earlier frontend (before the citation forms, the "the" rule, the compound split and the acronym rules), so these two rows are not strictly comparable with the others. Kaggle jobs: `fs2-12-prosody-en`, `fs2-14-extended`, `fs2-16-clarity`, `fs2-17-final-en`, `fs2-21-final-en-nostress`.

The final configuration has the lowest error over all 23 sentences, and it was preferred on listening over the earlier configurations. Removing the stress rule changes little (2.28 → 2.34 %); the rule is kept because it puts the homograph contrast of E2a/E2b into pitch (§6.1). One sentence is worse than with the plain model: the last clause of E4 ("so andaroo should arrive on time") is garbled for the ASR in every configuration with the current frontend (15.7–17.1 %), with or without the stress rule, whereas the model with only the post-filter and the earlier frontend gave "So, Antaro should arrive on time" (2.9 %). The cause was not isolated; the stretch is a made-up word followed by function words that the model predicts very short (§6.1, E4).

Model-only renders of the five English test sentences (no inference-time option and no post-filter; the same frontend) are in `Evaluation/English/model_only/` for comparison.

### 5.4 English

| ID | Duration | Frontend decisions | Whisper large-v3 transcript of the generated audio | CER raw / norm. |
|---|---|---|---|---|
| E1 | 8.65 s | "The" before a vowel → `DH IY0`; *antidisestablishmentarianism* is not in the lexicon → split into lexicon words anti + disestablishment + arianism (`AE1 N T AY0 D IH0 S IH0 S T AE1 B L IH0 SH M AH0 N T AE1 R IY0 AH0 N IH2 Z AH0 M`); *parliamentary* in its citation form (restored /t/); micro-pauses before *considerable* and *the parliamentary* | The anti-disestablishmentarianism movement gained considerable momentum during the parliamentary proceedings. | 0.0 % / 0.0 % |
| E2a | 5.53 s | *conduct* tagged noun (NN) → `K AA1 N D AH0 K T`; micro-pause before *the ceremony* | The orchestra's conduct during the ceremony was praised by the audience. | 0.0 % / 0.0 % |

| | | | | |
|---|---|---|---|---|
| E2b | 6.01 s | *conduct* tagged verb (VB) → `K AA0 N D AH1 K T`; micro-pauses before *an experiment* and *to evaluate* | The researchers will conduct an experiment to evaluate the proposed method. | 0.0 % / 0.0 % |
| E3 | 4.16 s | "rhyth m" joined for pronunciation → *rhythm* `R IH1 DH AH0 M` (text unchanged); micro-pause before *unexpectedly* | The rhythm of the orchestra changed unexpectedly. | 0.0 % / 0.0 % |
| E4 | 8.39 s | "ten forty five A M on twelve November twenty twenty six"; AM as letter names (durations × 1.6); *twenty* in its citation form (restored /t/); *andaroo* (Telugu అందరూ, "everyone") → `AE0 N D AA1 R UW0` | The meeting will start at 10.45 a.m. on 12 November 2026. Suhant Arashudra I. One time. | 47.3 % / 17.1 % |

**Observations.** Four of the five sentences are transcribed exactly (E4's raw CER is mostly digit formatting: "10.45", "12", "2026"). **E1**: the long rare word is recognised as the intended word; it needed the compound split and the slower long-word timing. **E2a/E2b**: the homograph pair is transcribed correctly, with the stress contrast in pitch (§6.1). **E3**: exact, including the joined *rhythm*. **E4**: the numbers, the date and *AM* are recognised; the last clause is not ("Suhant Arashudra I. One time."): *andaroo* is a Telugu word (అందరూ, "everyone") that the ASR cannot know, and the function words after it are short (§5.3). Across the five sentences the clear-speech options make the endings complete and the pauses quieter; the remaining audible weaknesses are short unstressed function words and a slightly breathy voice quality (§5.2).

**Found by inspecting the outputs.** The first full English run (same checkpoint) read E3 as `R AY1 DH` + the letter name `EH1 M` ("rythe em") and ended *andaroo* in `OW`. Both were caught in `frontend_trace.json` and fixed by general rules, covered by tests (§1.2); the outputs were regenerated with the unchanged checkpoint.

### 5.5 Telugu

| ID | Duration | Frontend decisions | Whisper large-v3 transcript of the generated audio | CER | CER, script-mapped |
|---|---|---|---|---|---|
| T1 | 5.27 s | all words by rule; `ny` before `c` in విశ్లేషించారు | Neuro-Pharmacological పరిశోధనలో నూతన ఉషధాల ప్రభావాన్ని విస్లేశించారు. | 38.7 % | 38.7 % |
| T2a | 2.46 s | పట్టు → `p a tt tt u` | తాడు పయతని పట్టు చాలా బలంగా ఉంది. | 7.1 % | 7.1 % |
| T2b | 2.69 s | పట్టు → `p a tt tt u` (same as T2a) | ఆమేపట్టు చిరను ధరించి వేడుకకు వచ్చింది. | 5.9 % | 5.9 % |
| T3 | 2.14 s | the split అడ విలో gives the phones of అడవిలో; no pause between the parts | పులి అడవిలో నిస్సబ్ధంగా నడిచింది. | 6.9 % | 6.9 % |
| T4 | 9.13 s | 2026 → రెండు వేల ఇరవై ఆరు; 12వ → పన్నెండవ; 10:45 → పది నలబై ఐదు | 2026 నవం బర్పన్ నండవతేది ఉదయం 10.45 గంటలకు ప్రారంబమోతుంది సమయానికి సిద్ధంగా ఉంచండి | 38.8 % | 38.8 % |

**Observations.** All five sentences pass the frontend with every number expanded to Telugu words. Whisper large-v3's transcripts show what is recognised, with its weak Telugu in mind (copy-synthesis ceiling 33.1 %, §7): **T1**: the English loanword is recognised and written in Latin script (*Neuro-Pharmacological*), which is most of the CER; the rest is శ → స in పరిశోధనలో and విశ్లేషించారు and ఔ → ఉ in ఔషధాల. **T2a/T2b**: పట్టు is transcribed correctly in both senses (written and pronounced alike; only the prosody differs, §6.2); the errors are word-boundary merges (తాడు పయతని for తాడుపై అతని, ఆమేపట్టు) and vowel length (ఆమె → ఆమే, చీర → చిర). **T3**: the split అడ విలో is heard as the single word అడవిలో, i.e. joined as a listener would join it; the only error is again శ → స. **T4**: Whisper writes the spoken numbers back as digits (2026, 10.45), which the CER counts as errors, drops the opening ఈ సమావేశం, merges నవంబర్ పన్నెండవ తేదీ across word boundaries (the syllables are present) and hears ప్రారంభ without aspiration; the final ఉంచండి is exact.

Across all 23 Telugu sentences scored (§5.5, §7.3, §7.4) the same confusions recur: శ → స in every word that contains శ, ణ → న, and lost aspiration (కథ → కద, పాఠకులు → పాదకులు). The frontend keeps all these phones distinct, so the merger is in the acoustic model, in Whisper, or both; with this ASR the two cannot be separated estimated.

## 6. Variance-adaptor plots, per test sentence

For each sentence, three plots (`Results/<Lang>/variance_adaptor_plots/`): **duration** — frames per phone (bars; right axis in ms); **pitch** — phone-level F0 in Hz (steps over each phone's time span) with the F0 that WORLD extracts from the generated wav (grey dots; unvoiced frames left as gaps), showing whether decoder and vocoder realise the prediction; **energy** — phone-level energy (STFT-frame L2 norm, arbitrary units). There is no ground truth for test sentences. For English, where inference-time options are used, each plot shows **both** the model's own prediction (hollow circles on the duration plot, dashed black steps on pitch and energy) and the final values used for the audio; the analysis below describes the model's prediction first and then what the options changed. The numbers behind every plot are in `variance_raw/<ID>.json` (per-word figures: `tools/word_stats.py --source model|final`).

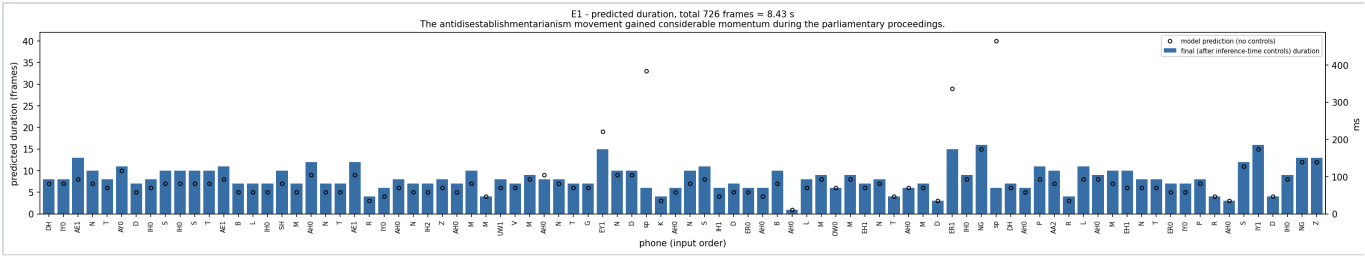
**Patterns common to all ten sentences (the model's own predictions)** measured

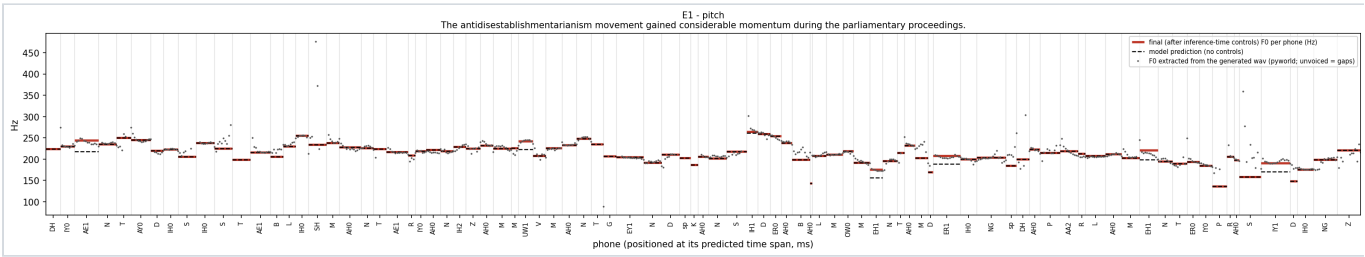
- **Declination**: the mean pitch of the last third of the phones is below that of the first third in every sentence (English 219–237 Hz → 195–209 Hz; Telugu 217–246 → 178–197 Hz). The exception to a falling end is E3, which the model ends on its highest pitch (269 Hz on *unexpectedly*), a question-like rise on a statement.
- **Phrase-final lengthening**: the last vowel is 1.1–2.5× the sentence's mean vowel duration in English (except E2a, 0.9×, whose last word ends in /s/) and 1.3–2.1× in Telugu.
- **Segmental energy pattern**: vowels and nasals high, stops, fricatives and pauses low (e.g. E4: /s/ ≈ 5–13, vowels 44–84 a.u.), as expected for an STFT-norm energy.
- **Speaking rate** 11.0–12.5 phones/s (English), 12.5–14.5 phones/s (Telugu).
- **Lexical stress (English)** is weakly marked: the noun and verb *conduct* get almost the same syllable durations (§6.1). This is what the data contain: in the Svarah training recordings the stressed vowel is the most prominent vowel of its word in 43.6 % of words, against 40.7 % expected by chance, and the model's predictions match the validation recordings (45.6 % vs 44.4 %) (`fastspeech2/stress_stats.py`, Kaggle `fs2-15-stress-stats`) measured.
- The F0 measured on the generated audio tracks the predicted phone-level F0 within voiced phones, so the decoder and vocoder realise the predicted pitch; deviations appear at phone boundaries and on short phones, as expected from step-wise phone-level targets.

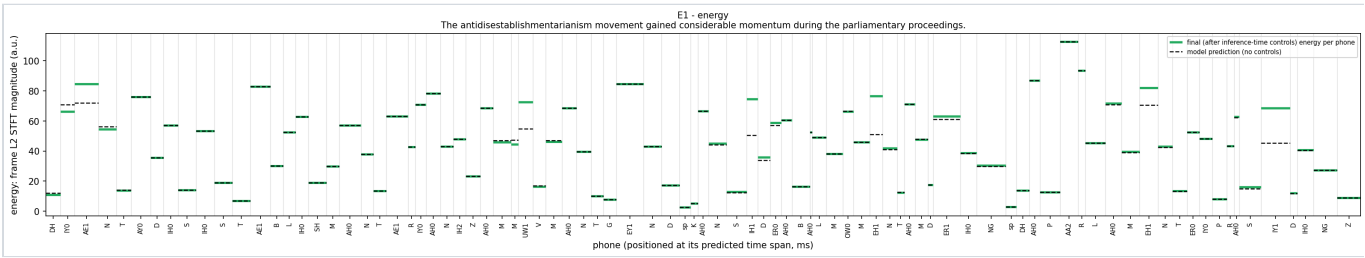
### 6.1 English

**E1 — The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.**

- **Duration (model).** *antidisestablishmentarianism* takes 2,020 ms (26 % of the model's 7.74 s) with uniformly short phones; *parliamentary* 801 ms, *proceedings* 882 ms. The final vowel is only 1.1× the mean vowel: little phrase-final lengthening. At the two micro-pause positions the model predicts 383 and 464 ms, i.e. it treats every pause symbol as a phrase break; the option sets them to 70 ms.
- **Pitch (model).** Several similar peaks (*considerable* 261, *antidisestablishmentarianism* 256, *movement* 248 Hz), lowest on *during* (204 Hz), first third 225 → last third 195 Hz: a flat, list-like contour with declination.
- **Energy (model).** Vowels median 63, nasals 43, other consonants 15 a.u., pauses ≈ 3; the final vowel of *proceedings* is 41.
- **Options.** Long words × 1.25 and everything × 1.10: the long word 2,020 → 2,798 ms, *parliamentary* 801 → 1,103 ms; the stress rule raised the stressed vowel of seven words (pitch up to × 1.12, energy up to × 1.5). Whisper recognises the long word exactly.

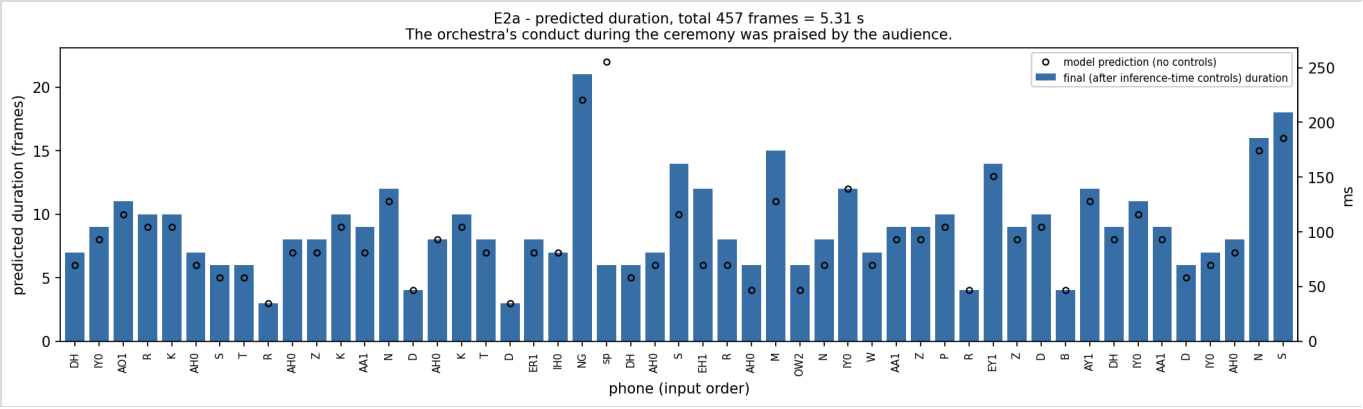


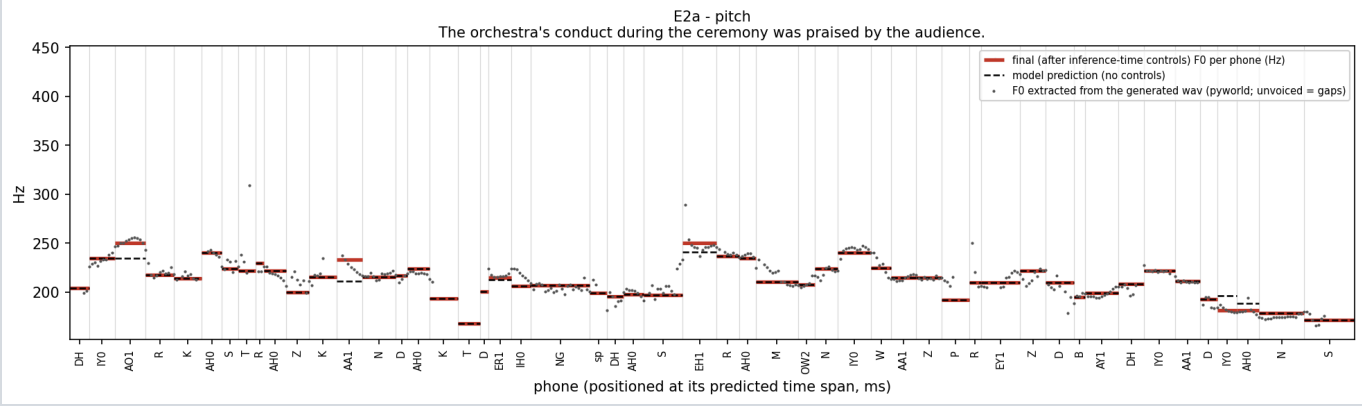



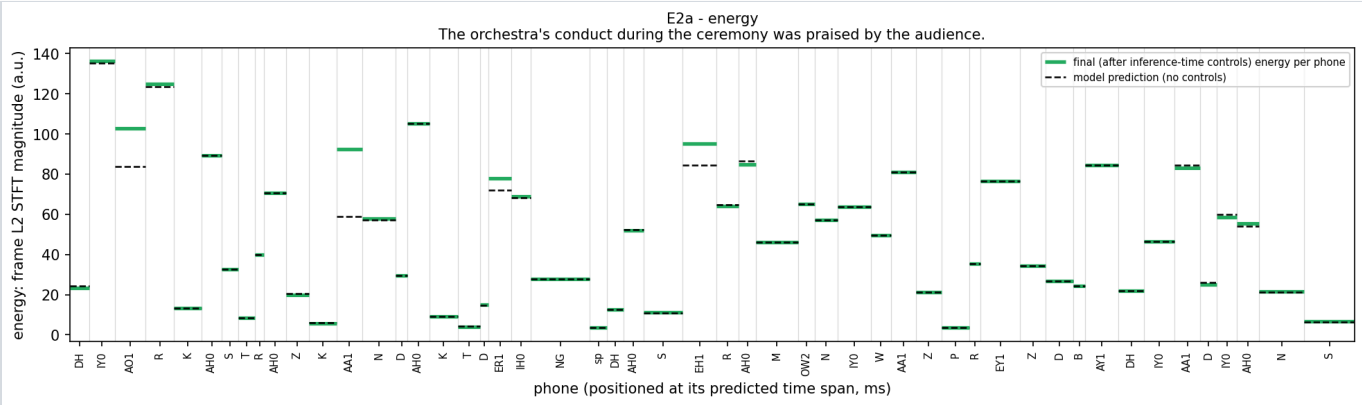




**E2a — The orchestra's conduct during the ceremony was praised by the audience.**

- **Duration (model).** *conduct* (noun, `K AA1 N D AH0 K T`): syllables 313 and 325 ms (ratio 0.96), the longest word of the sentence (639 ms). *audience* is long (662 ms) but its final vowel is short (0.9× the mean) because the word ends in /s/.
- **Pitch (model).** The stressed first vowel is *lower* than the unstressed second (211 vs 224 Hz): the model does not mark the noun's stress, in line with the data (§6, patterns). Main accents on *orchestra's* and *ceremony* (241 Hz); first third 220 → last third 201 Hz.
- **Energy (model).** Vowels median 74, nasals 46, other consonants 22 a.u.; highest on the /iː/ of the article before *orchestra's* (135).
- **Options.** The stress rule makes the first vowel of *conduct* the higher (233 vs 224 Hz) and the louder (× 1.5), with syllables 360 / 348 ms; the terminal fall lowers the last two phones to 182 Hz; the micro-pause before *the ceremony* is 70 ms (model: 255 ms).

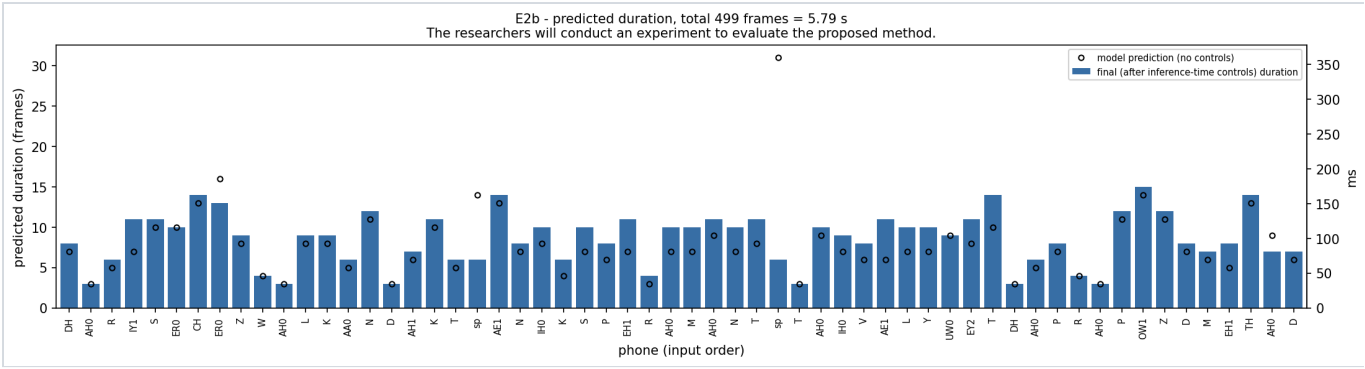


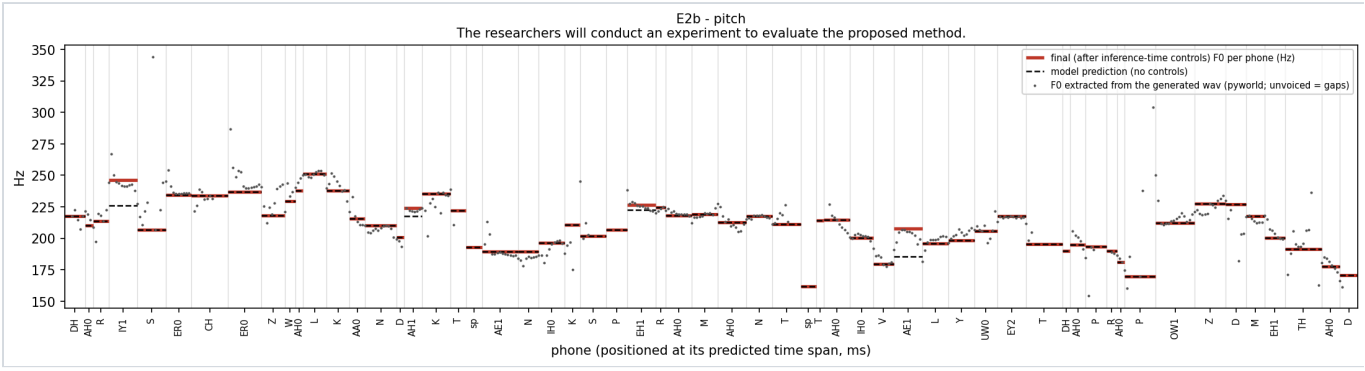



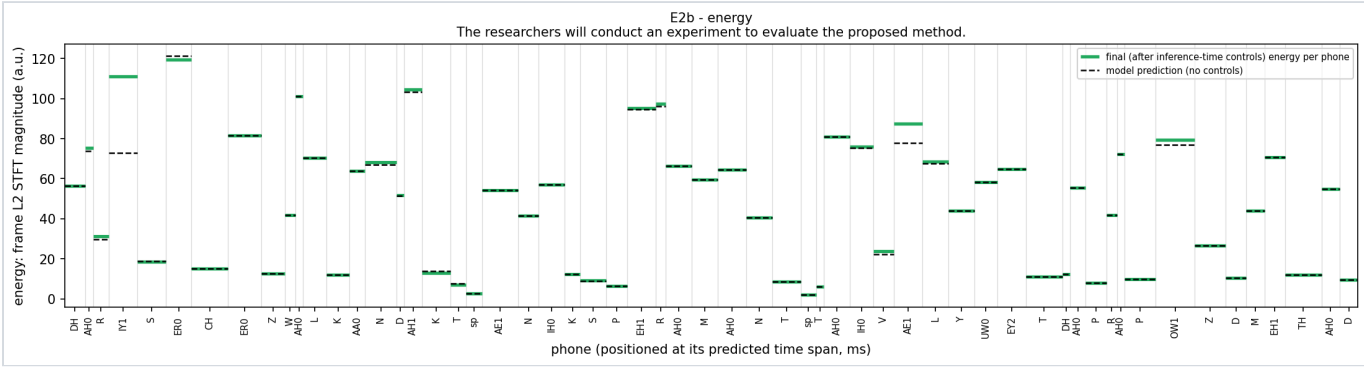




**E2b — The researchers will conduct an experiment to evaluate the proposed method.**

- **Duration (model).** *conduct* (verb, `K AA0 N D AH1 K T`): syllables 279 / 279 ms (ratio 1.00, against 0.96 for the noun) and 557 ms in all (noun: 639 ms). In the model's prediction the noun/verb difference is mainly the word's length, not which syllable is prominent. Final lengthening 1.2× on *method*.
- **Pitch (model).** Vowels of *conduct* 215 / 218 Hz (almost level); peaks on *will* (251 Hz) and *researchers* (237 Hz); lowest on *an* (189 Hz); first third 224 → last third 198 Hz.
- **Energy (model).** Vowels median 72, nasals 44, other consonants 12 a.u.; the strongest phone is the /ɚ/ of *researchers* (121).
- **Options.** The stress rule puts the higher pitch on the stressed second vowel (224 vs 215 Hz). With E2a (233 vs 224 Hz on the first vowel) the final audio therefore carries the homograph contrast in pitch, which the model alone hardly does. Whisper: 10.9 % CER without the options ("an experimental evaluation of"), exact with them.

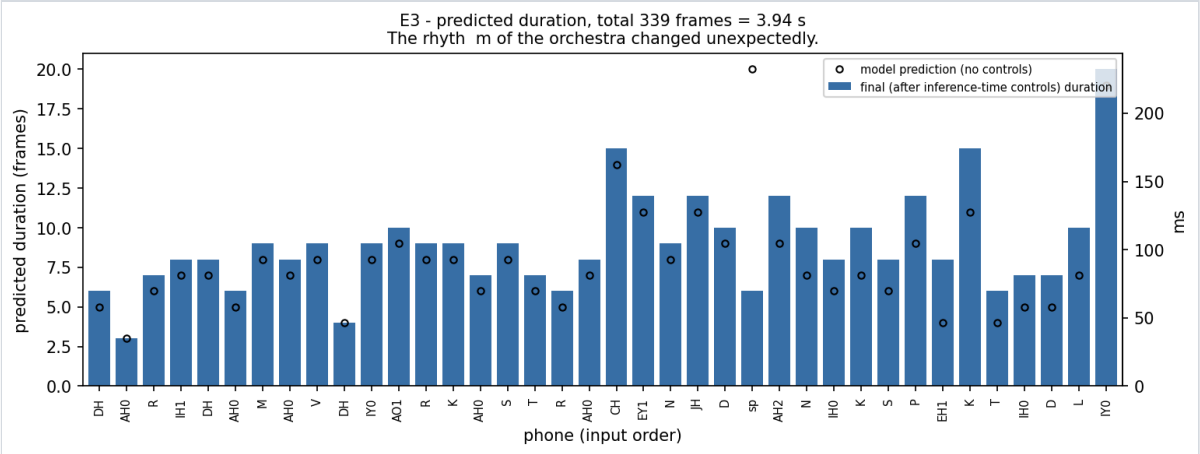


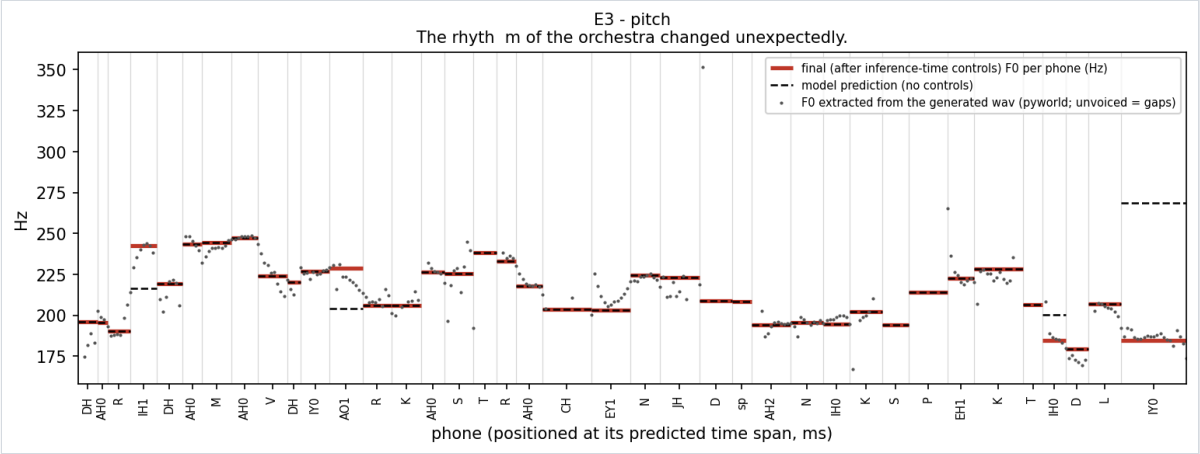



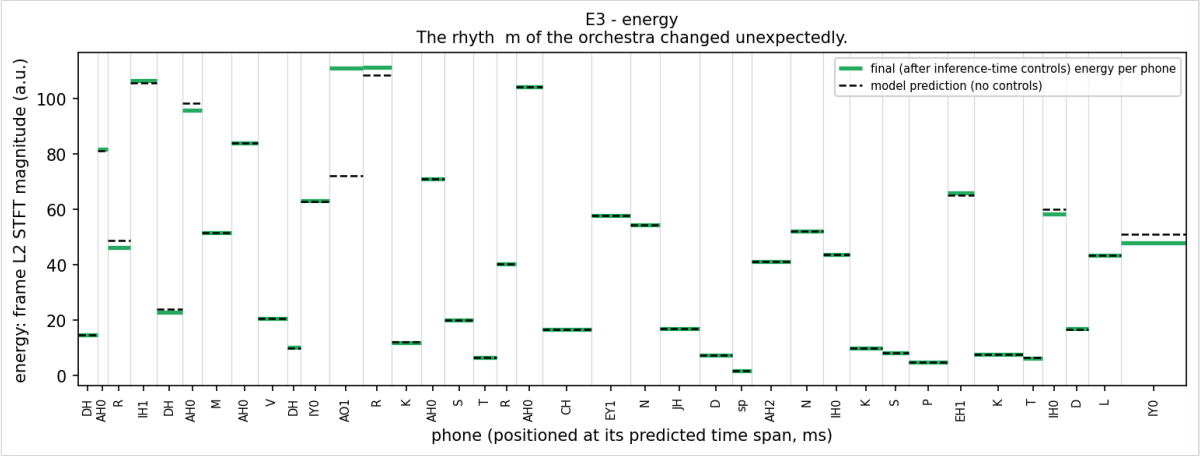




### E3 — The rhyth m of the orchestra changed unexpectedly.

- **Duration (model).** The split word is pronounced as *rhythm* (`R IH1 DH AH0 M`, 383 ms), without a pause inside it: the double space has no effect after joining. *unexpectedly* is very long (1,149 ms) with a final vowel of 221 ms (2.5× the mean), the strongest final lengthening of the five sentences.
- **Pitch (model).** The model ends this statement on its highest pitch: the last vowel of *unexpectedly* is 269 Hz, a question-like rise, while *rhythm* and *of* carry 244–248 Hz.
- **Energy (model).** Vowels median 68, nasals 52, other consonants 16 a.u.; the /r/ of *rhythm* is the energy maximum (108), as an approximant with vowel-like energy.
- **Options.** The terminal-fall rule lowers the last two vowels from 201 and 269 Hz to 185 Hz, turning the rise into a statement fall; *unexpectedly* 1,149 → 1,544 ms (long word); the stress rule adds energy to the first vowel of *orchestra* (× 1.5).

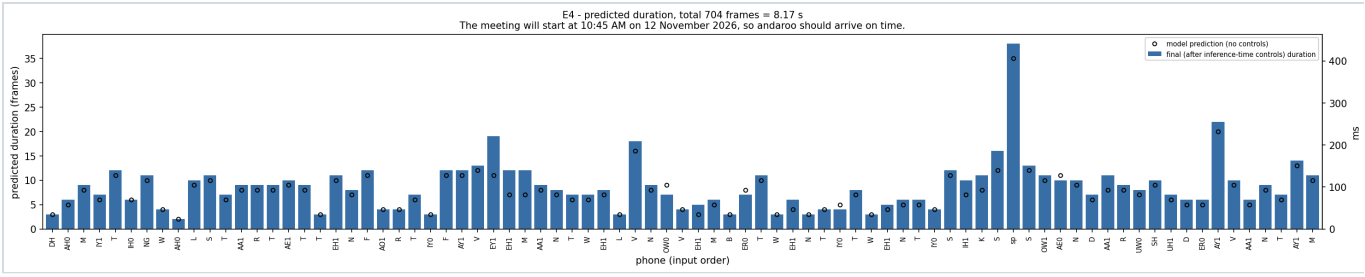


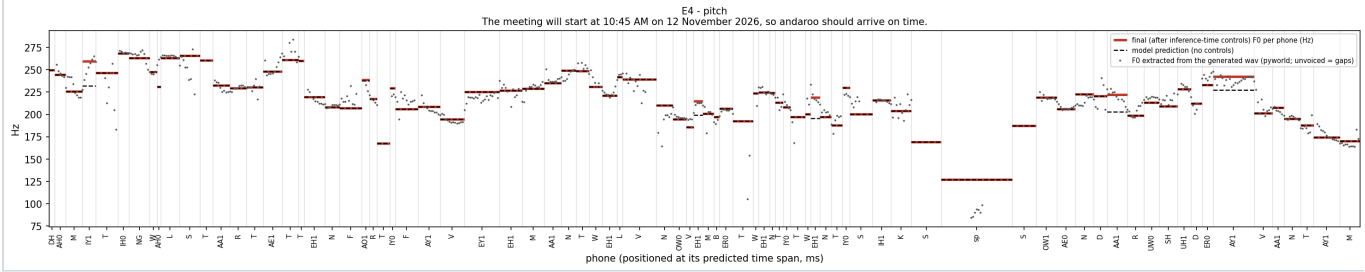



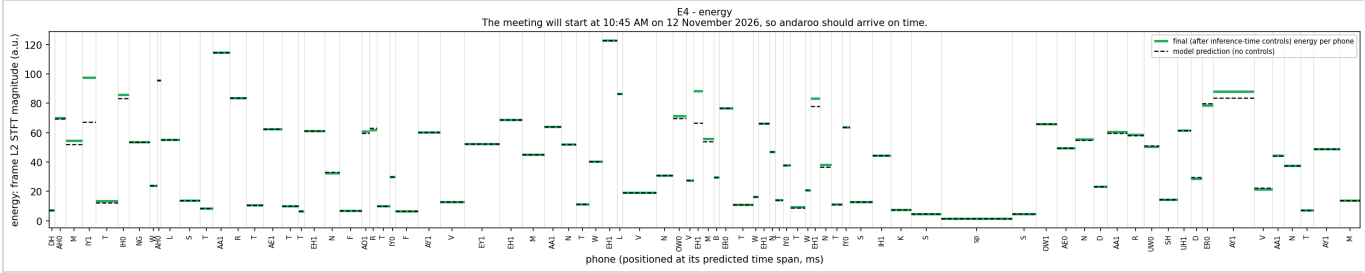




**E4 — The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.**

- **Duration (model).** Numbers read as words with growing durations (*ten* 232, *forty* 325, *five* 395 ms; *twelve* 441, *November* 464, *six* 441 ms): each number group ends long, list-like; *twenty twenty* 337 + 325 ms; *AM* 290 ms. The comma gives a 406 ms pause. After it the function words are very short (*should* 232 ms with a 70 ms vowel, *on* 151 ms); final lengthening 1.8× on *time*.
- **Pitch (model).** Highest on *meeting* (268 Hz), lowest on *time* (188 Hz); no pitch reset after the comma (*six* 216 → *so* 219 Hz), unlike Telugu T4; first third 237 → last third 205 Hz.
- **Energy (model).** Vowels median 64, nasals 46, other consonants 13 a.u.; /s/ of *six* and *so* ≈ 5 (fricatives), the comma pause 1.4.
- **Options.** *AM* as letter names × 1.6 (× 1.10): 290 → 499 ms; comma pause 441 ms; *twenty* in its citation form with /t/; the stress rule lengthens the stressed vowels of *November* (× 1.4), the first *twenty* (× 1.31) and *andaroo* (× 1.24). The last clause remains the weak point for the ASR (§5.4).


E4 - predicted duration, total 704 frames = 8.17 s
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.
model prediction (no controls)
final (after inference-time controls) duration
predicted duration (frames)
ms
phone (input order)

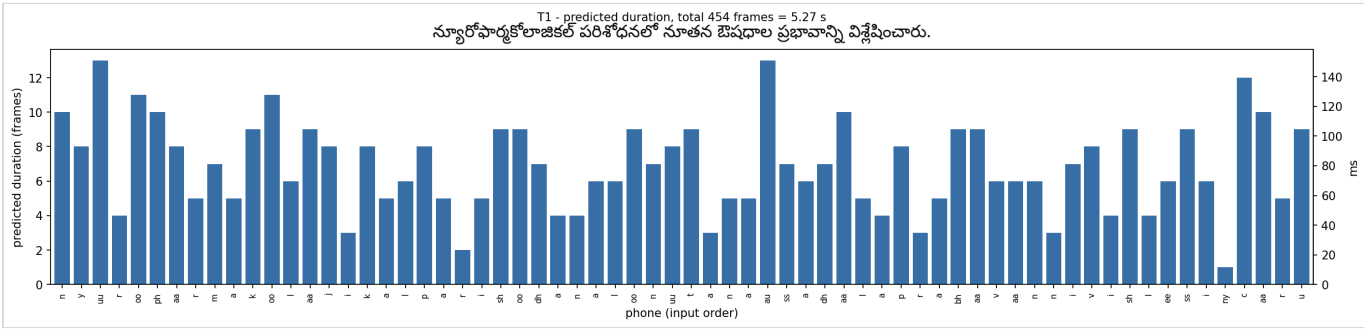

E4 - pitch
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.
final (after inference-time controls) F0 per phone (Hz)
model prediction (no controls)
F0 extracted from the generated wav (pyworld; unvoiced = gaps)
Hz
phone (positioned at its predicted time span, ms)

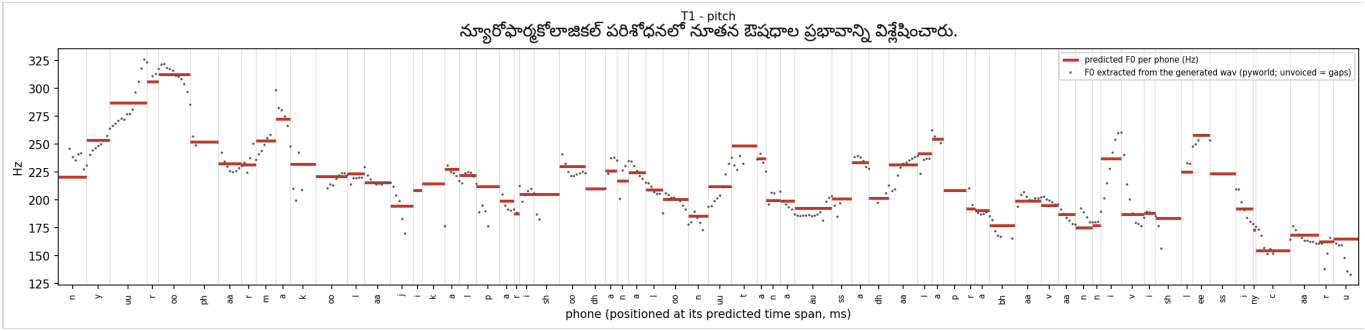

E4 - energy
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.
final (after inference-time controls) energy per phone
model prediction (no controls)
energy: frame L2 STFT magnitude (a.u.)
phone (positioned at its predicted time span, ms)

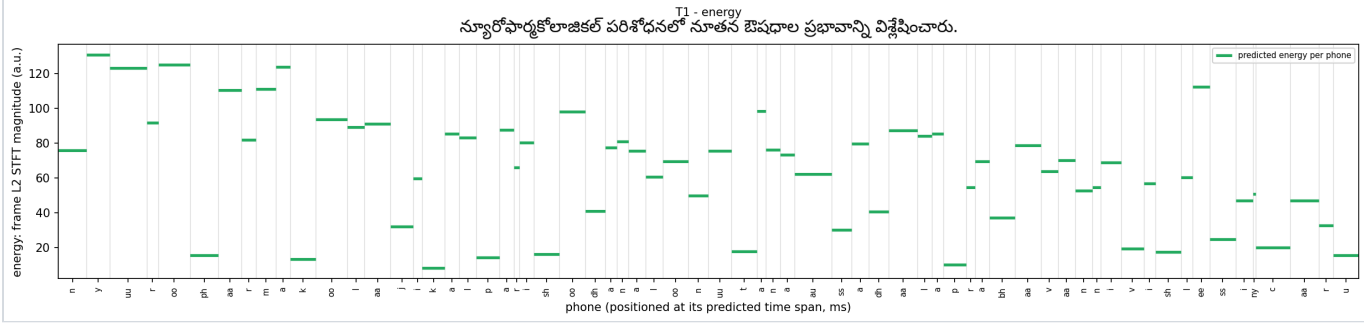
### 6.2 Telugu

#### T1 — న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

- న్యూరోఫార్మకోలాజికల్ lasts 1,695 ms (32 % of 5.27 s) and carries the sentence's highest pitch (313 Hz): the long loanword is the focus of the sentence-initial phrase.
- Pitch drops to 230 Hz on పరిశోధనలో and stays in a narrow 237–258 Hz band for the rest: a typical Telugu (verb-final) declarative pattern with the main prominence early.
- The final verb విశ్లేషించారు is long (964 ms; cluster `sh 1`, nasal `ny` before `c`) with 1.3× final lengthening; no internal pauses (no internal punctuation).
- **Energy.** Vowels median 79, nasals 65, other consonants 35; the final /u/ of విశ్లేషించారు falls to 15: the energy fades over the last syllables, as in a declarative ending.

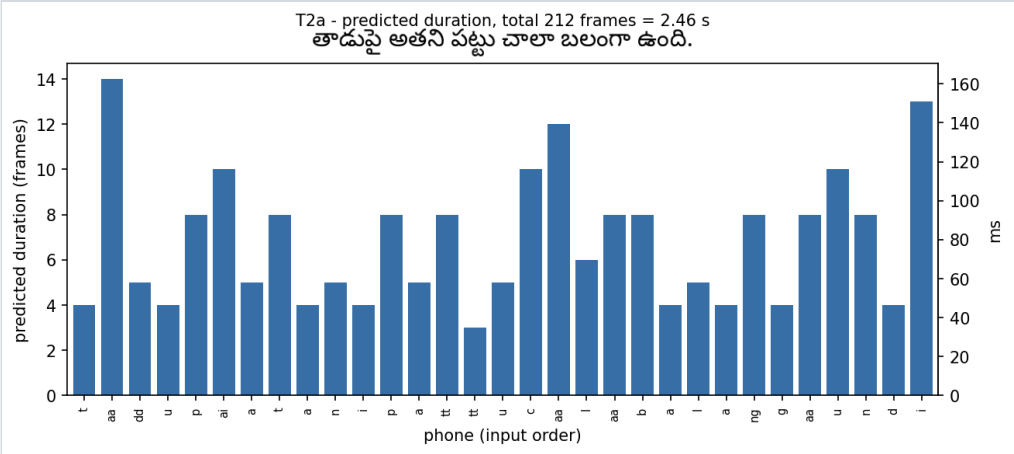


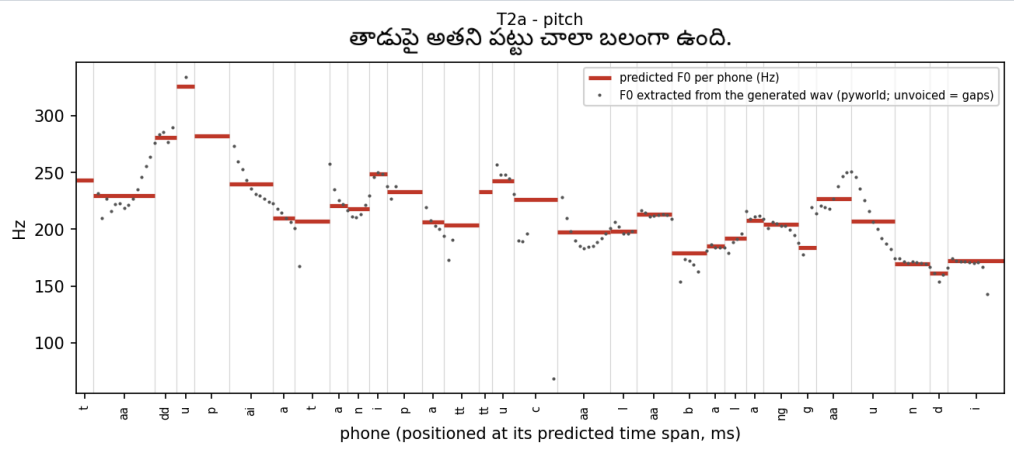



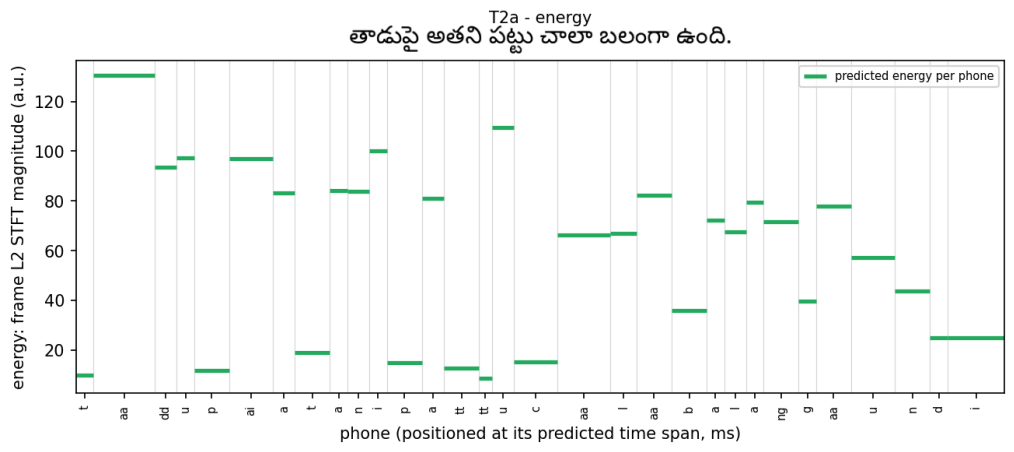




#### T2a — తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

- Peak pitch on the first word తాడుపై (326 Hz), then a steady fall to ఉంది (207 Hz).
- పట్టు (`p a tt tt u`, geminate retroflex) lasts 337 ms with 242 Hz, inside the falling line, not accented.
- Final vowel 151 ms, 1.8× the mean: strong phrase-final lengthening.
- **Energy.** Vowels median 82, nasals 72, other consonants 19; the geminate `tt tt` of పట్టు is the energy minimum (8.6), the long closure of the stop; the final /i/ drops to 25.

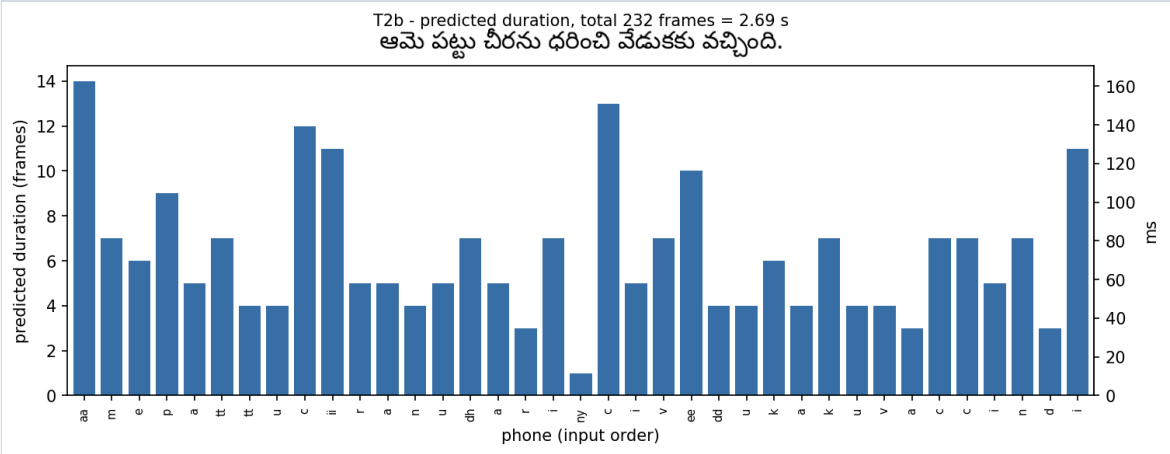


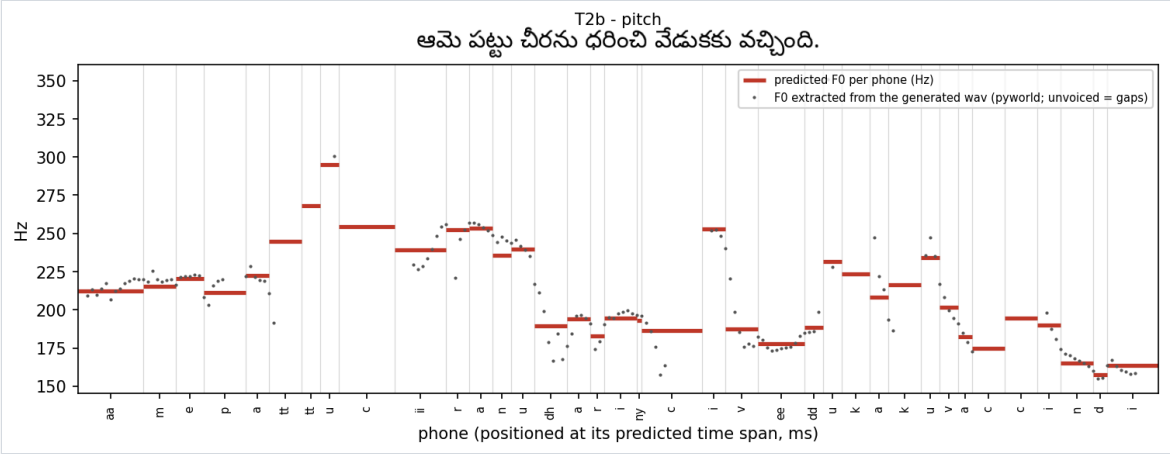



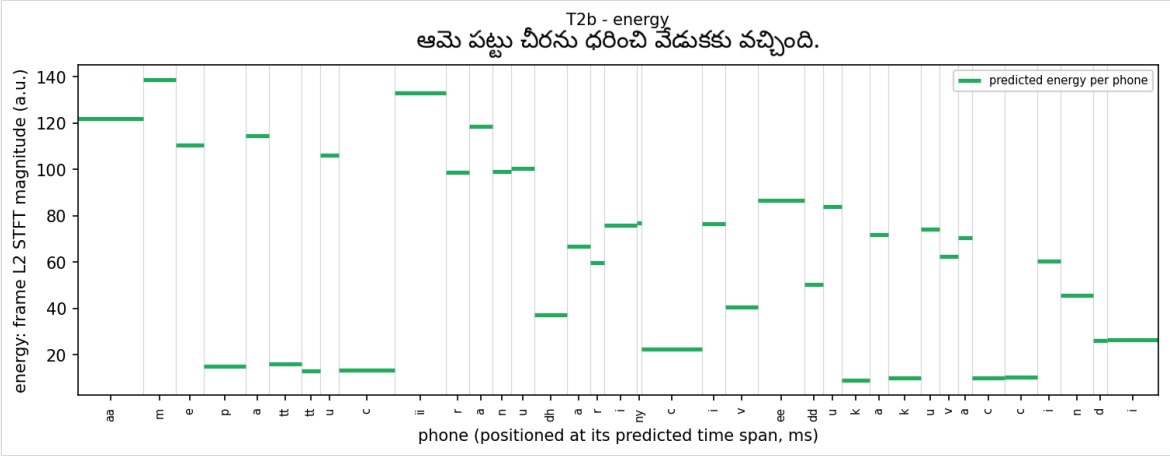




### T2b — ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

- పట్టు has the same phones and the same duration (337 ms) as in T2a, as it should (the two words are written and pronounced alike), but now carries the sentence's pitch peak (295 Hz vs 242 Hz in T2a): as the modifier in పట్టు చీర (silk saree) it is accented, after a low start on ఆమె (220 Hz).
- The rest declines from 254 Hz (చీరను) to 202 Hz (వచ్చింది), with 1.7× final lengthening.
- The difference between T2a and T2b is therefore purely prosodic (position and focus), which the model reproduces.
- **Energy.** Vowels median 84, nasals 88: the /m/ of ఆమె is the strongest phone (139), so the sentence starts loud; other consonants 19; the final /i/ drops to 26.

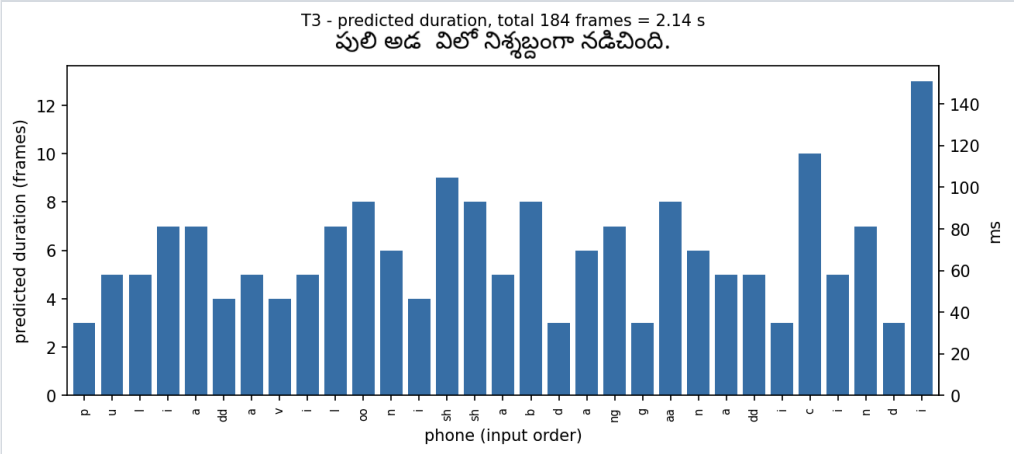


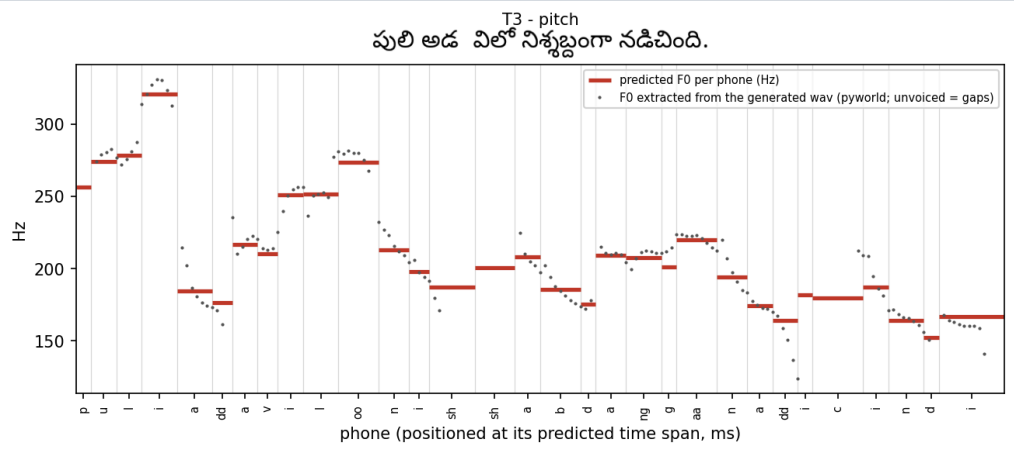



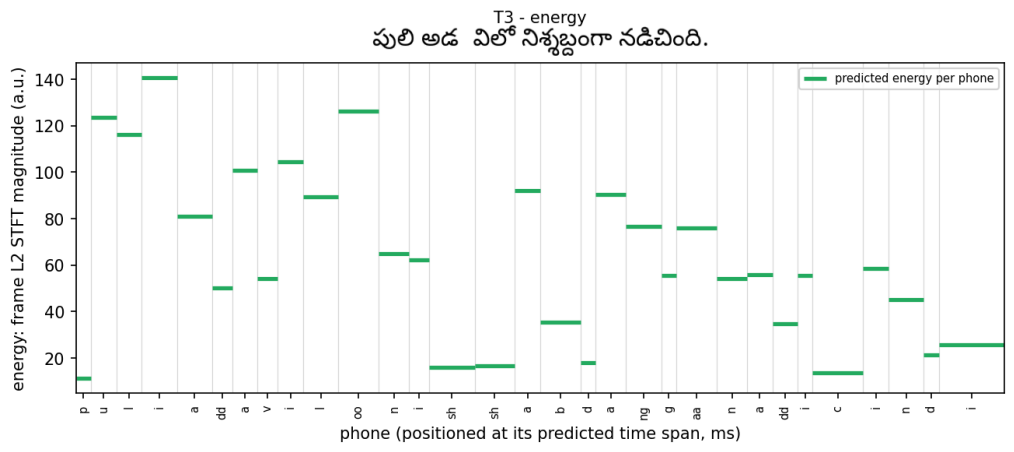




### T3 — పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

- Peak on పులి (321 Hz, the topic).
- The broken word అడ  విలో gives exactly the phone sequence of అడవిలో (`a dd a v i l oo`; checked), because the Telugu input has no word-boundary symbol: the model receives the intended word. It is spoken without a pause (అడ 186 ms, విలో 279 ms, contiguous), with pitch rising from 217 to 274 Hz across the word.
- నిశ్శబ్దంగా is long (778 ms; geminate `sh sh` and the cluster `b d`); నడిచింది ends the sentence at the pitch minimum (194 Hz) with 2.1× final lengthening.
- **Energy.** Vowels median 86, nasals 60, other consonants 35; strongest on an /i/ (141); the final /i/ of నడిచింది drops to 26.

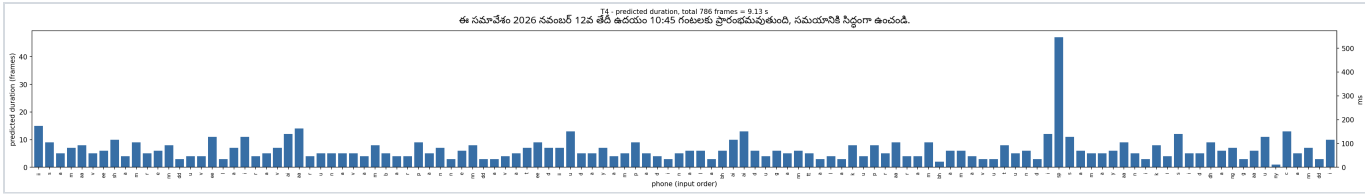


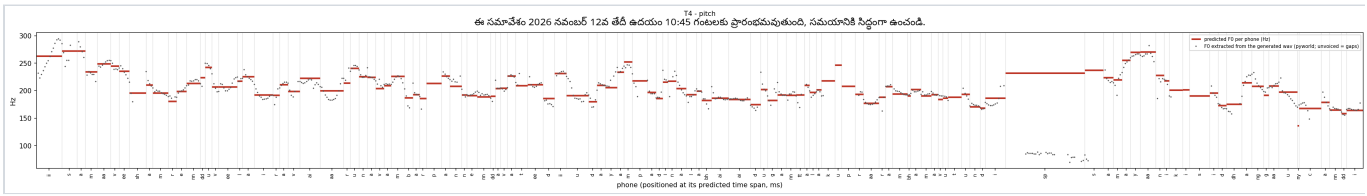



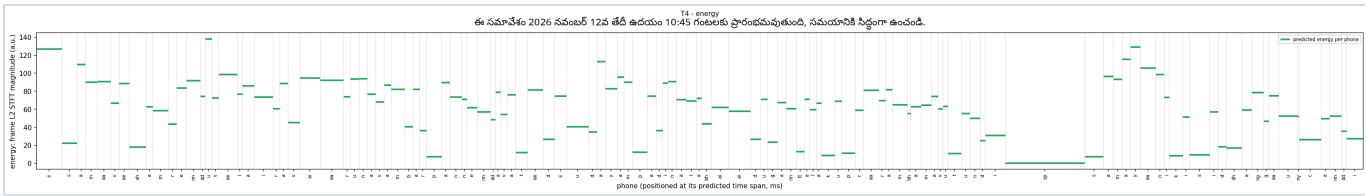




**T4 — ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.**

- Longest sentence (9.13 s, 120 phones). Numbers are expanded to Telugu words: రెండు వేల ఇరవై ఆరు (302/290/453/267 ms), పన్నెండవ (615 ms), పది నలభై ఐదు (244/418/267 ms), read without hesitation pauses.
- The comma gives a 546 ms pause followed by a clear **pitch reset**: from ≈ 208 Hz on ప్రారంభమవుతుంది (the longest word, 1,126 ms) up to 270 Hz on సమయానికి: the second clause starts a new intonation phrase.
- Highest pitch on సమావేశం (272 Hz), lowest on the final ఉంచండి (198 Hz), with 1.5× final lengthening.
- **Energy.** Vowels median 75, nasals 74, other consonants 37; the comma pause is near silence (0.3); the final /i/ drops to 28. In all five Telugu sentences the last vowel has 15–28 against a vowel median of 75–86, a stronger end-of-sentence decay than in English (41–55 against 63–74).


T4 - predicted duration, total 786 frames = 9.13 s
ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.
predicted duration (frames)
ms
phone (input order)


T4 - pitch
ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.
Hz
predicted F0 per phone (Hz)
F0 extracted from the generated wav (pyworld; unvoiced = gaps)
phone (positioned at its predicted time span, ms)


T4 - energy
ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.
energy: frame L2 STFT magnitude (a.u.)
predicted energy per phone
phone (positioned at its predicted time span, ms)

## 7. Objective evaluation on held-out data beyond the brief

Both models were evaluated on **every validation utterance** (never trained on), each synthesised with its own speaker and without any inference-time option (`fastspeech2/evaluate_val.py`, Kaggle `fs2-04-eval`, `fs2-06-eval-fix`). Silence phones are excluded from duration/pitch/energy metrics.

| Metric measured | English (742 utt.) | Telugu (446 utt.) |
|---|---|---|
| Duration MAE (frames, 11.6 ms) / Pearson r of log(1+d) | 2.81 / 0.58 | 1.06 / 0.85 |
| Total utterance length, relative error | 11.7 % | 4.5 % |
| Phone-level pitch RMSE / Pearson r | 34.6 Hz / 0.35 | 29.2 Hz / 0.76 |
| Phone-level energy RMSE / Pearson r | 26.8 / 0.78 | 17.4 / 0.90 |
| Mel-cepstral distortion*, ground-truth duration/pitch/energy (DTW) | 36.6 | 26.1 |
| Mel-cepstral distortion*, predicted duration/pitch/energy (DTW) | 39.0 | 30.9 |
| Whisper large-v3 CER / WER, our synthesis (150 utt.) | 15.7 % / 21.5 % | 35.8 % / 80.3 % |
| Same, copy-synthesis ceiling (ground-truth mel → same HiFi-GAN) | 11.6 % / 13.8 % | 33.1 % / 78.6 % |

* Computed from the model's own 80-band log-mel (orthonormal DCT, c1–c24), averaged over DTW-aligned frames. This is not the WORLD/SPTK mel-cepstrum MCD reported in papers; it is comparable only within this table. CER/WER are corpus-level (total edits / total reference length) after lower-casing and removing punctuation; CER ignores spaces. Whisper large-v3 is weak on Telugu: its CER on copy-synthesis of real Telugu validation recordings is 33.1 %, against 11.6 % for English, so Telugu CERs bound pronunciation errors from above rather than measure them. A Telugu ASR model (`vasista22/whisper-telugu-large-v2`, not trained on IndicTTS; pinned in `evaluate_val.ASR_REVISIONS`) was prepared for this evaluation (Kaggle `fs2-20-asr-te-cpu`), but it runs on CPU (the weekly GPU quota was used up) and had not finished when this report was built.

**Reading the table.** (1) The variance predictors cost little spectral accuracy: replacing ground-truth duration, pitch and energy by predictions raises the distortion by 2.4 (en) and 4.8 (te); most of the remaining error is in the decoder itself. (2) Telugu prosody is predicted far better than English (pitch r 0.76 vs 0.35; duration r 0.85 vs 0.58): with two speakers and read studio speech, phone identity and context determine prosody much more than in 110-speaker, partly spontaneous Svarah. (3) For English the intelligibility gap to the vocoder ceiling is 4.1 points of CER: the acoustic model loses some intelligibility, but most of the error budget is shared with copy-synthesis (accented speech, ASR limits). (4) For Telugu, Whisper large-v3 loses nearly as much on copy-synthesis of real recordings (33.1 %) as on our synthesis (35.8 %): as far as this ASR can tell, the acoustic model costs 2.7 points of CER.

### 7.1 Speed measured

| Real-time factor (compute s / audio s), batch 1, warm | English | Telugu |
|---|---|---|
| GPU (Tesla T4): acoustic model / HiFi-GAN / total | 0.0079 / 0.0193 / 0.027 | 0.0070 / 0.0185 / 0.025 |
| CPU (4 threads, fp32): acoustic / HiFi-GAN / total | 0.029 / 0.322 / 0.350 | 0.026 / 0.355 / 0.381 |

On a T4 one second of speech takes ~26 ms (≈ 37–40× faster than real time); the paper reports RTF $1.95 \times 10^{-2}$ on a V100 with Parallel WaveGAN paper. On a CPU the acoustic model is still ~35× faster than real time; the vocoder dominates (~90 % of the time), so a lighter vocoder is the lever for CPU deployment. Timing excludes the text frontend; CUDA is synchronised before each reading; one untimed warm-up call.

### 7.2 Prosody controls measured

| Control (factor) | English: length / median F0 / RMS | Telugu: length / median F0 / RMS |
|---|---|---|
| none | 4.55 s / 206 Hz / 0.146 | 2.46 s / 209 Hz / 0.148 |
| duration 0.8 / 1.25 | 3.37 s / 5.42 s | 1.88 s / 3.02 s |
| pitch 0.8 / 1.2 (after M12) | 169 Hz / 245 Hz | 165 Hz / 252 Hz |

| | | |
|---|---|---|
| pitch 0.8 / 1.2 (reference code) | 140 Hz / 113 Hz — inverted (former voice, 126 Hz) | 204 Hz / 209 Hz — almost no effect |
| energy 0.8 / 1.2 (after M12) | RMS 0.126 / 0.164 | RMS 0.120 / 0.159 |

The duration control scales length close to the factor (0.74× and 1.19× for English; 0.76× and 1.23× for Telugu; rounding of per-phone frames explains the difference). Pitch control after M12 is near-proportional (×0.79/×1.20 for Telugu).

### 7.3 Stress sentences

Sixteen further sentences (`Evaluation/stress_sentences.json`) probe cases the test set does not: questions, homographs, names, abbreviations, numbers, conjuncts, code-mixing, very long and very short sentences. English uses the final configuration; Telugu the model as submitted.

**English (final configuration; CER after Whisper's English normaliser, against the original text): corpus 1.7 %**

| ID | Case | Text | Whisper large-v3 transcript | CER norm. |
|---|---|---|---|---|
| SE01 | time and numbers | The train leaves at 7:30 PM from platform 3. | The train leaves at 7.30 pm from Platform 3. | 0.0 % |
| SE02 | question intonation | Did you remember to bring the documents for tomorrow's interview? | Did you remember to bring the documents for tomorrow's interview? | 0.0 % |
| SE03 | homograph (wind: noun / verb) | The wind was strong, so we had to wind the rope tightly. | The wind was strong, so we had to wind the rope tightly. | 0.0 % |
| SE04 | Indian names and places | Priya and Venkatesh travelled from Hyderabad to Chennai last weekend. | Priya and Venkatesh travelled from Hyderabad to Chennai last weekend. | 0.0 % |
| SE05 | rare technical words | The cryptographic protocol relies on elliptic curve signatures. | The cryptographic protocol relies on elliptical signatures. | 9.1 % |
| SE06 | abbreviations and acronyms | Dr. Rao presented the results at IIIT Hyderabad. | Dr. Lau presented the results at IIT Hyderabad. | 7.0 % |
| SE07 | long sentence | Although the monsoon arrived later than expected this year, the farmers in the eastern districts managed to sow their crops on time, thanks to the new irrigation canals that were completed in early spring. | Although the monsoon arrived later than expected this year, the farmers in the eastern districts managed to sow their crops on time, thanks to the new irrigation canals that were completed in early spring. | 0.0 % |
| SE08 | exclamation, very short | What a wonderful surprise! | What a wonderful surprise! | 0.0 % |

SE05: "elliptic curve" heard as "elliptical"; SE06: the spelled acronym IIIT heard as "IIT" and *Rao* as "Lau".

**Telugu (CER against the normalised text): corpus 18.9 %**

| ID | Case | Text | Whisper large-v3 transcript | CER, script-mapped |
|---|---|---|---|---|
| ST01 | question intonation | మీరు రేపు ఉదయం ఆఫీసుకు వస్తారా?<br>Will you come to the office tomorrow morning? | మీరు రేపు ఉదైయమాఫి సుకువస్తారా. | 15.4 % |
| ST02 | numbers | ఈ పుస్తకం ధర 250 రూపాయలు.<br>The price of this book is 250 rupees. | ఈ పుస్తకం ధర రెండు వందల యాభి రూపాయలు. | 6.7 % |
| ST03 | time | రైలు సాయంత్రం 6:30 కి బయలుదేరుతుంది.<br>The train departs at 6:30 in the evening. | రేలు సాయంత్రమారు ముప్పైకి బయలు దేరుతుంది. | 8.3 % |

| | | | | |
|---|---|---|---|---|
| ST04 | consonant clusters (conjuncts) | స్త్రీ శక్తి రాష్ట్ర అభివృద్ధికి ముఖ్యం.<br>Women's strength is important for the state's development. | 3. సక్తి రాష్ట్ర ఆభివరుద్ధి కి ముఖ్యమ్ | 37.1 % |
| ST05 | English loanwords in Telugu script (with ZWNJ) | నా లాప్‌టాప్ బ్యాటరీ త్వరగా అయిపోతోంది.<br>My laptop battery is draining quickly. | 4 లాప్టాప్ బ్యాటరి త్వరగా ఆయి పోతోంది. | 12.1 % |
| ST06 | Latin-script word inside Telugu (code-mixing) | నేను నిన్న కొత్త iPhone కొన్నాను.<br>I bought a new iPhone yesterday. | నేను నిన్న కొత్త ఆఈ ఫోన్ కొన్నాను. | 21.4 % |
| ST07 | long sentence | హైదరాబాద్ నగరంలో ప్రతి సంవత్సరం జరిగే పుస్తక ప్రదర్శనకు వేలాది మంది పాఠకులు, రచయితలు మరియు విద్యార్థులు ఉత్సాహంగా హాజరవుతారు.<br>Thousands of readers, writers and students enthusiastically attend the book fair held every year in Hyderabad. | AIDARABADNAGARAN లో ప్రతి సంవ్రచం జరిగే పుస్తక ప్రదర్శనకు వేలాది మంది పాదకులు, రచైతలు మరియు విద్యార్థులు ఉత్సాహంగా హాజరవుతారు. | 20.2 % |
| ST08 | exclamation, very short | ఎంత అందమైన దృశ్యం!<br>What a beautiful sight! | ఎంతా అందమైన ద్రుస్యం. | 33.3 % |

ST06: the Latin-script *iPhone* is transcribed ఆఈ ఫోన్, i.e. spoken as intended through `text/telugu/latin.py`. Whisper writes digits where it hears none ("3." for స్త్రీ in ST04, "4" for నా in ST05) and drops question and exclamation marks (ST01, ST08), which the CER ignores.

### 7.4 Extended sentence set

Ten more sentences per language (`Evaluation/Extended/extended_sentences.json`) extend each test sentence's difficulty (long technical words, homographs, numbers and dates, clusters, questions and exclamations, long sentences); some are variants of the test sentences, and several were used while developing the frontend rules (§1.2), so they are not a blind test.

**English: corpus CER 2.1 %**

| ID | Case | Text | Whisper large-v3 transcript | CER norm. |
|---|---|---|---|---|
| XE01 | Long, uncommon words | The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings. | The anti-disestablishmentarianism movement gained considerable momentum during the parliamentary proceedings. | 0.0 % |
| XE02 | Homographs, stress | The orchestra's conduct during the ceremony was praised, while the researchers will conduct another experiment tomorrow. | Bay Orchestra's conducting the ceremony was praised, while the researchers will conduct another experiment tomorrow. | 5.8 % |
| XE03 | Sibilants, articulation | She sells seashells beside the seashore, but the shells she sells are surely not the same. | She sells seashells beside the seashore, but the shells she sells are surely not the same. | 0.0 % |
| XE04 | Numbers, time, dates | At 10:45 AM on November 12, 2026, approximately 1,247 researchers will gather for the international conference. | At 10.45 a.m. on November 12, 2026, approximately 1,247 researchers will gather for the international conference. | 0.0 % |
| XE05 | Consonant clusters, pronunciation | The rhythmic patterns of the extraordinarily talented musician changed unexpectedly during the performance. | The rhythmic patterns of the extraordinarily talented musician changed unexpectedly during the performance. | 0.0 % |

| | | | | |
|---|---|---|---|---|
| XE06 | Abbreviations, decimals | Dr. Williams asked whether the temperature would reach 37.5 degrees Celsius by 8:30 p.m. | Dr Williams asked whether the temperature would reach 37.5°C by 8.30pm. | 17.6 % |
| XE07 | Long sentence, prosody | Although artificial intelligence can process millions of calculations per second, human creativity remains remarkably difficult to replicate. | Although artificial intelligence can process millions of calculations per second, human creativity remains remarkably difficult to replicate. | 0.0 % |
| XE08 | Homographs, tense-dependent pronunciation | I read the book yesterday, and today I will read another chapter before going to sleep. | I read the book yesterday and today I will read another chapter before going to sleep. | 0.0 % |
| XE09 | Questions, emotional intonation | Wait! Are you seriously suggesting that we should rebuild the entire system from scratch? | Wait, are you seriously suggesting that we should rebuild the entire system from scratch? | 0.0 % |
| XE10 | Phonetic coverage, fluency | The quick brown fox jumps over the lazy dog while five extraordinary musicians quietly rehearse their final symphony. | The quick brown fox jumps over the lazy dog while five extraordinary musicians quietly rehearse their final symphony. | 0.0 % |

XE06's remaining CER is formatting: the ASR writes "37.5°C", which its normaliser does not map to "degrees Celsius"; the content is right. XE02: "The orchestra's conduct during" heard as "Bay Orchestra's conducting". Audio: `Evaluation/Extended/English/`.

**Telugu: corpus CER 17.1 %**

| ID | Case | Text | Whisper large-v3 transcript | CER, script-mapped |
|---|---|---|---|---|
| XT01 | Long Sanskrit-derived technical vocabulary | న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు. | Neuro-Pharmacological పరిసోధనలో నూతన ఉషధాల ప్రభావాన్ని విస్లేశించారు. | 38.7 % |
| XT02 | Homographs: పట్టు | తాడుపై అతని పట్టు చాలా బలంగా ఉంది, కానీ ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది. | తాడు పయతని పట్టు చాలా బలంగా ఉంది, కాని ఆమే పట్టు చిరను ధరించి వేడుకకు ఉచ్చింది. | 9.1 % |
| XT03 | Consonant combinations, long sentence | పులి అడవిలో నిశ్శబ్దంగా నడిచింది, కానీ దాని గర్జనతో అడవి మొత్తం ఒక్కసారిగా ఉలిక్కిపడింది. | పులి అడవిలో నిస్చబ్దంగా నడిచింది. కాని దాని గర్జనతో అడవి మొత్తం మొక్క సారిగా ఉలిక్కి పడింది. | 6.5 % |
| XT04 | Dates, numbers, time | ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది. | 2026 నవం బర్వన్నేండవతేది ఉదైయం 10.45 గంటలకు ప్రారంభమోతుంది | 50.0 % |
| XT05 | క్ష, శ, త్ర combinations | క్షణక్షణానికి మారుతున్న పరిస్థితులను గమనిస్తూ శాస్త్రవేత్తలు పరిశోధనను కొనసాగించారు. | క్షణక్షనానికి మారుతున్న పరిస్తితులను గమనిస్తు సాస్త్ర వేత్తలు పరిసోధన అనుకొన సాగించారు. [Kannada script, mapped] | 7.8 % |
| XT06 | Compound words, pronunciation | హైదరాబాద్‌లోని విశ్వవిద్యాలయంలో కంప్యూటర్ సైన్స్ విద్యార్థులు కృత్రిమ మేధస్సుపై చర్చించారు. | AIDARABAD లోని విస్వవిద్యాలైయం లోకం కూప్యూటర్ సైంస్ విద్యార్థులు కృత్రిమమే ధస్సుపై చర్చించారు. | 19.5 % |
| XT07 | Natural conversational Telugu | అమ్మ చెప్పిన కథలోని చిన్నారి ప్రతిరోజూ ఉదయాన్నే పక్షుల కిలకిలారావాలను వింటూ పెరిగింది. | అమ్మా చెప్పిన కదలోని చిన్నారి ప్రతి రోజు ఉదైయాన్నే పక్షులకి లక్కిలా రావాలను వింటు పెరిగింది. | 9.2 % |
| XT08 | Prosody, sentence rhythm | వర్షం పడుతున్నప్పటికీ, రేపటి ప్రయాణం కోసం మేము అవసరమైన వస్తువులన్నింటినీ సిద్ధం చేసుకున్నాము. | వర్షం పడుతున్నప్పటికి రేపటి ప్రయానం కోసం మేము అవసరమైన వస్తువుల అన్నింటి ని సిద్ధం చేసుకున్నాము. | 4.9 % |

| | | | | |
|---|---|---|---|---|
| XT09 | Question intonation, emphasis | నిజంగా మనం సాధించాలనుకున్న లక్ష్యాన్ని చేరుకోవడానికి క్రమశిక్షణ, సహనం, పట్టుదల అవసరం కాదా? | నిజంగా మనం సాధించాలను కున్న లక్షాన్ని చేరుకోవడానికి క్రమ సిక్షన, సహనం, పట్టుదల అవసరం కాదా. [Kannada script, mapped] | 5.1 % |
| XT10 | Repeated syllables, long-form pronunciation | శ్రీకాకుళం నుంచి తిరుపతికి వెళ్లే రైలులో ప్రయాణిస్తున్న ప్రయాణికులకు రేపు ఉదయం ప్రత్యేక ప్రకటన చేయబడుతుంది. | స్రికా కుళ్ళమ్ను౦చి తిరుపతికి వెల్లె రైలుల్లో ప్రయానిస్తున్న ప్రయానికులక్కు రేపు ఉదైయం ప్రత్తెక్క ప్రకటన చేయవడుత్తుంది. [Kannada script, mapped] | 23.2 % |

Whisper wrote XT05, XT09 and XT10 in Kannada script (76–107 Kannada characters each; raw CER 100–113 %); the table scores them after mapping the Kannada block back onto the Telugu block (same layout, offset 0x80; `tools/build_report.py`). XT09's question particle కాదా is recognised; its intonation falls (§8.2). XT04 and XT01 repeat T4's and T1's ASR effects (digits, Latin script). Whisper transcripts: `Evaluation/Telugu/metrics_extended_whisper.json` (Kaggle `fs2-14-extended`).

## 8. English vs Telugu; limitations, failures and improvements

### 8.1 Comparison

| | English | Telugu |
|---|---|---|
| Data character | 110 speakers, ~2.7 min each, Indian-accented, varied recording and style | 2 speakers, ~2.5 h each, studio read speech |
| Text → phones | irregular spelling: lexicon + neural G2P; homographs, out-of-vocabulary words and normalisation are the main risks | near-phonemic script: deterministic rules cover every word; risks are normalisation and loanwords |
| Prosody prediction (validation) | weak: pitch r 0.35, duration r 0.58; lexical stress barely marked, as in the data | good: pitch r 0.76, duration r 0.85 |
| Over-fitting | mild (validation drifts up after 7k) | none visible (flat after ~10k) |
| Output quality without inference-time help | over-smoothed decoder, recording noise reproduced, endings cut; needs the post-filter and clear-speech options (§5.2–5.3) | not over-smoothed; used as the model produces it |
| Phrase structure | pause at the E4 comma with no pitch reset after it in the model's prediction | pause at the T4 comma followed by a clear pitch reset (208 → 270 Hz) |
| Main failures on the test set | E4's last clause (a made-up word followed by very short function words) is garbled for the ASR; short unstressed function words; lexical stress, question rises and statement falls come from rules, not from the model (§5.3, §6); the muffled first test voice (§5.1) | శ heard as స, ఐ as న, lost aspiration (model or ASR, not separable here); word-boundary merges; T4's spoken numbers written back as digits by the ASR; no question intonation (XT09, §8.2) |

### 8.2 Limitations

- **5 h of data** per language (the assignment's budget) against 24 h in the paper; the variance predictors memorise the training set (§4) and English prosody generalises poorly.
- **No listening test**: quality statements rely on ASR and acoustic measurements.
- **English decoder over-smoothing** (§5.2): the GV post-filter restores the spectral variance but only part of the frame-to-frame sharpness; the remaining smoothness is a property of the trained decoder on 110 short-duration speakers.
- **The English output relies on inference-time options** (§5.3); the model alone (`Evaluation/English/model_only/`) is less clear. They are rules, not learned: they fire on punctuation and word length, not on meaning.
- **No question intonation in Telugu**: the Telugu model, used without rules, ends the yes/no question XT09 (… అవసరం కాదా?) with a falling, fading final vowel (175 Hz against a sentence median of 208 Hz), i.e. with statement intonation; the "?" is not part of the model input.
- **Telugu G2P is orthographic**: it transcribes the script; it does not model schwa reduction, loanword pronunciations (ఫ్ as /f/), or sandhi. It was not reviewed by a native speaker.
- **Universal vocoder**: copy-synthesis costs 11.6 % CER on English validation speech; recording quality varies by Svarah speaker and bounds what each voice can sound like.
- **Telugu clips that MFA could not align were dropped** (28); their transcripts were not inspected.
- **English validation shares speakers with training** (split by recording, §2.1).

### 8.3 Possible improvements (from the test sentences)

- **Voice and data selection by recording quality** (§5.1): select or weight training speakers by measured clarity, not only by amount of audio; the test-voice choice alone removed most English ASR errors.

- **Word-boundary and sentence-type symbols in the input**: neither frontend marks word boundaries or the sentence type (as in the reference), so the model cannot separate "అడ విలో" from "అడవిలో", cannot learn the rise of a question (XT09) and ends E3 with a rise; adding such symbols needs retraining estimated.
- **Early stopping / checkpoint selection** on validation loss (English ~7–10k steps).
- **Against over-smoothing** (largest remaining English gap): a decoder loss that does not average (mixture output or an adversarial post-net), which needs retraining; or a longer vocoder fine-tune that starts from trained discriminators estimated. Speaker adaptation needs more than 2.7 min of the target voice, or parameter-efficient adaptation (conditional layer norm, as in AdaSpeech).
- **Learned instead of rule-based clarity**: training on clean, single-speaker English data of the same size would remove the need for denoising and edge handling; slower, clearer speech could be learned from data with that style.
- **Telugu**: native-speaker review of the G2P rules; question and exclamation marks as input symbols; frame-level pitch with CWT (as in the paper) for richer intonation.

## 9. Reproducibility

**Software** (as run on Kaggle): Python 3.13, PyTorch 2.11.0 (CUDA 12.8 on T4; CPU build for data jobs), librosa, pyworld, tgt, g2p_en, MFA 3.4.3 (conda-forge, via micromamba), transformers 5.16.1 (evaluation only). Full package lists are in each job's `environment.txt`; `requirements.txt` lists the minimum versions. Data: Svarah and IndicTTS Telugu from Hugging Face at pinned revisions (`data/ingest.py`); ASR models at pinned revisions (`evaluate_val.ASR_REVISIONS`).

**Configuration**: `English/config.yaml`, `Telugu/config.yaml` (preprocess / model / train sections; step values as used), and the exact configs as run in `Results/<Lang>/train_config_used.yaml`.

**Commands** (data root `D`; Telugu shown, then the English lines that differ):

```
cd fastspeech2
python -m data.ingest --lang te --cache /tmp/hf --out D        # needs a Hugging Face token
python -m data.select_split --manifest_dir D/manifests/te --train_hours 5 --val_hours 1
python data/mfa_prepare.py --lang te --ingest_dir D --corpus /tmp/corpus_te \
       --dict_out D/mfa_te/dictionary.txt
mfa train /tmp/corpus_te D/mfa_te/dictionary.txt D/mfa_te/model.zip \
       --output_directory D/preprocessed/te/TextGrid
# English: mfa align /tmp/corpus_en D/mfa_en/dictionary.txt english_us_arpa \
#          D/preprocessed/en/TextGrid --beam 100 --retry_beam 400
cd ..
python Telugu/preprocess.py --data_root D
python Telugu/train.py      --data_root D --max_minutes 480
python Telugu/inference.py  --data_root D --checkpoint Telugu/checkpoints/te_model.pth.tar \
       --speaker te_female --out_dir Results/Telugu
python fastspeech2/evaluate_val.py --config Telugu/config.yaml --data_root D \
       --checkpoint Telugu/checkpoints/te_model.pth.tar --speaker te_female --out_dir Evaluation/Telugu

# English (pretrained MFA aligner above; 210-minute budget = 17,949 steps)
python English/preprocess.py --data_root D
python English/train.py      --data_root D --max_minutes 210
python tools/noise_profile.py --config English/config.yaml --data_root D \
       --checkpoint English/checkpoints/en_model.pth.tar --speaker svarah_8a8e2655 \
       --gv_ratio Results/English/gv_ratio.json --gv_strength 1.0 --out Results/English/noise_profile.json
python English/inference.py  --data_root D --checkpoint English/checkpoints/en_model.pth.tar \
       --speaker svarah_8a8e2655 --gv_ratio Results/English/gv_ratio.json --gv_strength 1.0 \
       --prosody_fall --prosody_stress 1.0 --prosody_phrasing --duration_control 1.10 \
       --long_word_scale 1.25 --heavy_word_pause --denoise_profile Results/English/noise_profile.json \
       --edge_fade --out_dir Results/English
# gv_ratio.json: fastspeech2/postfilter_eval.py (Kaggle fs2-08-postfilter). Without the options: the model alone.
```

**Checkpoints**: GitHub release v1.0 (`en_model.pth.tar`, `te_model.pth.tar`; SHA-256 in `<Lang>/checkpoints/README.md`). Inference alone needs only a checkpoint and the `stats.json` / `speakers.json` in

`Results/<Lang>/`. **Kaggle jobs** (`kaggle/`, listed in `kaggle/README.md`) reproduce every stage, and each records the commit it ran. **Tests**: `pytest`.

## Appendix: deliverables checklist

| Assignment item | Where |
|---|---|
| Two separate models, English (Svarah) and Telugu (IndicTTS Telugu), ~5 h train / ~1 h validation each | §2.1, §3; `English/config.yaml`, `Telugu/config.yaml` |
| Data selection and subsampling strategy; train/validation separate with no overlap | §2.1; `fastspeech2/data/select_split.py`; `Evaluation/Data/accounting.json` |
| Dataset preparation and preprocessing, both languages | §2.2 |
| Changes to the GitHub implementation for each language, particularly Telugu | §1.2–1.3; `MODIFICATIONS.md` ("Telugu support at a glance", L1–L4, M1–M12, new files) |
| Speech for all test sentences, text preserved as provided | `Results/<Lang>/test_generated_audios/` (E1, E2a, E2b, E3, E4; T1, T2a, T2b, T3, T4); §5 (text check) |
| Folder structure: `English/`, `Telugu/` (train.py, preprocess.py, inference.py, config.yaml, checkpoints/), `Results/`, `Report/`, `README.md`, `requirements.txt` | repository root |
| Trained checkpoints, separately per language | GitHub release v1.0, linked from `<Lang>/checkpoints/README.md` and `README.md` (files exceed GitHub's 100 MB limit) |
| Training- and validation-loss plots, both models | `Results/<Lang>/training_loss.png`, `validation_loss.png`; §4 |
| Variance-adaptor plots (duration, pitch, energy) for each test sentence, both languages | `Results/<Lang>/variance_adaptor_plots/{duration,pitch,energy}/`; §6 |
| Report: implementation, architecture, modifications | §1 |
| Report: data, subsampling, preprocessing, split, training configuration | §2, §3 |
| Report: loss curves, convergence, instability, over-fitting | §4 |
| Report: generated speech per test sentence, pronunciation, quality | §5.4, §5.5 |
| Report: variance-adaptor analysis per test sentence | §6 |
| Report: English vs Telugu, limitations, failures, improvements | §8 |
| Report: reproducibility (dependencies, configs, commands) | §9; `README.md` |